

Graduate Quantum Mechanics

Minimum Review and Exam Equations

Course lecture extracts: Chapters 1–4

Purpose. A compact, section-mapped review for a reader who has studied graduate quantum mechanics before. Retain the equations needed to reconstruct results, the assumptions under which they apply, and the distinctions most likely to cause mistakes. Exam-useful means useful for solving problems in this material; it is not a prediction of the instructor’s examination.

Scope. Covers lecture pages 1–100: `extract1.pdf` pp. 1–57, the Chapter 3 continuation pp. 58–74, and `lecture-note-SQS.pdf` Chapter 4, lecture pp. 75–100 (PDF file pages 85–110). Includes Chapters 1–3 and all of Chapter 4 (4.1–4.8), with problem roadmaps. Source-page labels refer to printed lecture page numbers, not PDF file positions.

How to use. First identify the problem’s Hilbert space, basis, observable or Hamiltonian, and assumptions. Then select the relevant formula, perform the calculation, and check normalization, units, limiting cases, and phases. Attempt each self-check before consulting its answer. Small historical subsections are grouped, with original section numbers retained.

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Chapter 1: The Birth of Quantum Mechanics—Fundamental Equations

Minimum review of all lecture subsections 1.1–1.12, including 1.5.1. Source equations are retained in the lecture’s conventions; paragraphs marked **Supplement** add reconstruction steps or safeguards, not extra lecture claims. This is a review guide, not a prediction of examination questions.

1.1, 1.11, 1.12 Historical frame and structural synthesis

Source PDF pp. 1, 12–13.

Keep the structure, not the biographies. With $\hbar = h/(2\pi)$, action becomes phase in units of $\hbar$. The old quantum theory supplied discrete spectra and quantized action but no general consistent dynamics. By the 1927 Solvay meeting, wave and matrix mechanics had consolidated into equivalent descriptions. Correct equations often preceded their interpretations (Planck quanta, Born probabilities, Dirac antiparticles); the later path integral added another formulation.

Structural step	What must be reconstructed
Energy/momentum and waves	$E = \hbar\omega$, $\mathbf{p} = \hbar\mathbf{k}$
Noncommutative kinematics	$[\hat{q}, \hat{p}] = i\hbar$
Dynamics versus measurement	Schrödinger evolution versus Born probabilities
Representation independence	Matrices and wave functions describe the same state
Relativity / classical limit	Spinors / stationary action in the path integral

Operators emphasize algebra and symmetry; wave functions emphasize boundary-value problems; paths emphasize spacetime and semiclassical reasoning. No single equation alone supplies the full theory.

1.2 Planck’s law and the appearance of $\hbar$

Source PDF pp. 2–3.

For equilibrium electromagnetic radiation, $u_\nu(T) d\nu$ is energy per volume in $[\nu, \nu + d\nu]$. With c the speed of light and $k_B = R/N_A$,

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/(k_B T)} - 1}, \quad \bar{E} = \frac{h\nu}{e^{h\nu/(k_B T)} - 1}.$$

Reconstruct: count mode excitations $E_n = nh\nu$, $n = 0, 1, \dots$, weight by $e^{-nh\nu/(k_B T)}$, and multiply $\bar{E}$ by mode density $8\pi\nu^2/c^3$. For $x = h\nu/(k_B T)$,

$$x \ll 1: \quad u_\nu \simeq \frac{8\pi\nu^2}{c^3} k_B T \quad (\text{Rayleigh–Jeans}), \quad x \gg 1: \quad u_\nu \simeq \frac{8\pi h\nu^3}{c^3} e^{-x} \quad (\text{Wien}).$$

Supplement—derivation/pitfalls: $Z = \sum_{n \geq 0} e^{-nx} = (1 - e^{-x})^{-1}$, so $\bar{n} = -\partial_x \ln Z = (e^x - 1)^{-1}$. This is excitation energy, not zero-point energy. u_ν is per ordinary frequency, not per wavelength or angular frequency; changing variable requires a Jacobian. Extending Rayleigh–Jeans to all ν makes $\int_0^\infty u_\nu d\nu$ diverge; the exponential tail removes this.

1.3 Planck–Einstein and de Broglie relations

Source PDF pp. 3–4.

$$E = h\nu = \hbar\omega, \quad \mathbf{p} = \hbar\mathbf{k}, \quad p = h/\lambda, \quad (E/c, \mathbf{p}) = \hbar(\omega/c, \mathbf{k}).$$

The photoelectric relation is $K_{\max} = h\nu - W$, with work function W . The plane wave $\psi \propto e^{i(\mathbf{k}\cdot\mathbf{r} - \omega t)} = e^{i(\mathbf{p}\cdot\mathbf{r} - Et)/\hbar}$ connects energy and momentum to phase gradients. A single-valued orbital phase motivates the old semiclassical condition $\oint p_i dq_i = n_i h$ (one action cycle; not an implicit sum). **Supplement:** emission requires $\nu \geq W/h$; a negative computed $K_{\max}$ means no photoemission. Distinguish ν from $\omega = 2\pi\nu$.

1.4 The old quantum theory as a bridge

Source PDF pp. 4.

$$E_n = -\frac{\text{Ry}}{n^2}, \quad n = 1, 2, \dots, \quad \text{Ry} \simeq 13.6 \text{ eV}, \quad h\nu = E_{n_i} - E_{n_f} = \text{Ry} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right).$$

These are nonrelativistic hydrogen energies; emission has $n_i > n_f$ and positive photon energy. The successful spectrum did not validate classical electron orbits as a general dynamics. **Supplement:** $\oint p_i dq_i = n_i h$ is the lecture's old quantization rule, not a universal exact condition (modern semiclassical rules can require turning-point phase corrections).

1.5 Matrix mechanics: noncommuting observables

Source PDF pp. 5–6.

Position and momentum are operators, not simultaneously assigned classical numbers. For canonical variables and a Hamiltonian $\hat{H}$,

$$\boxed{[\hat{q}, \hat{p}] = i\hbar \hat{I}}, \quad \boxed{\frac{d\hat{A}}{dt} = \frac{i}{\hbar} [\hat{H}, \hat{A}] + \left(\frac{\partial \hat{A}}{\partial t} \right)_{\text{explicit}}}.$$

The latter is the Heisenberg-picture equation; in particular, $\dot{\hat{q}} = (i/\hbar)[\hat{H}, \hat{q}]$ and $\dot{\hat{p}} = (i/\hbar)[\hat{H}, \hat{p}]$ when neither has explicit time dependence. **Sign check:** $(i/\hbar)[\hat{H}, \hat{A}] = [\hat{A}, \hat{H}]/(i\hbar)$; reversing the commutator without changing the prefactor is wrong.

Supplement—reconstruct Hamilton-like motion. Use $[AB, C] = A[B, C] + [A, C]B$. For $\hat{H} = \hat{p}^2/(2m) + V(\hat{q})$,

$$[\hat{p}^2, \hat{q}] = -2i\hbar \hat{p}, \quad [V(\hat{q}), \hat{p}] = i\hbar V'(\hat{q}) \implies \dot{\hat{q}} = \hat{p}/m, \quad \dot{\hat{p}} = -V'(\hat{q}).$$

These manipulations assume a common suitable operator domain and sufficient regularity of V . Commuting with $\hat{H}$ implies conservation only if the explicit time derivative also vanishes.

1.5.1 Why the commutator precedes uncertainty

For Hermitian observables in a normalized state with finite variances,

$$(\Delta A)^2 = \langle \hat{A}^2 \rangle - \langle \hat{A} \rangle^2, \quad \boxed{\Delta A \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|} \implies \Delta q \Delta p \geq \hbar/2.$$

The modulus is essential: the commutator expectation is generally imaginary. This concerns the spread of measurement outcomes in a state, not experimental clumsiness. **Supplement—proof backbone:** center the operators, $\delta A = \hat{A} - \langle \hat{A} \rangle$, $\delta B = \hat{B} - \langle \hat{B} \rangle$. Cauchy–Schwarz gives $\Delta A \Delta B \geq |\langle \delta A \delta B \rangle| \geq |\text{Im} \langle \delta A \delta B \rangle| = \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$. A zero commutator expectation makes this lower bound zero; it need not make both variances zero or make the operators commute.

1.6 Wave mechanics: the Schrödinger equation

Source PDF pp. 7.

For one nonrelativistic particle in a scalar potential,

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}), \quad \boxed{i\hbar \partial_t \psi(\mathbf{r}, t) = \hat{H} \psi(\mathbf{r}, t)}, \quad \hat{H} \psi(\mathbf{r}) = E \psi(\mathbf{r}).$$

Acting on the plane wave motivates $E \mapsto i\hbar \partial_t$ and $\mathbf{p} \mapsto -i\hbar \nabla$. Applying these to $E = p^2/(2m) + V$ motivates, but does *not derive*, the new dynamical postulate from classical mechanics. Spectra become differential-operator eigenvalue problems.

Supplement—assumptions/link: when $\hat{H}$ is time independent, write $\Psi_E(\mathbf{r}, t) = \psi_E(\mathbf{r}) e^{-iEt/\hbar}$. This converts the evolution equation into $\hat{H} \psi_E = E \psi_E$; a general state is a superposition, not one energy eigenstate. Self-adjoint $\hat{H}$ with appropriate boundary conditions generates norm-preserving unitary evolution. For time-independent $\hat{H}$, $U(t, t_0) = e^{-i\hat{H}(t-t_0)/\hbar}$. Do not use this simple exponential blindly for a time-dependent noncommuting Hamiltonian. The displayed $\hat{H}$ omits spin, relativistic effects, and magnetic vector potentials.

1.7 One quantum mechanics, many representations

Source PDF pp. 7–8.

In a complete orthonormal basis,

$$|\psi\rangle = \sum_n c_n |n\rangle, \quad c_n = \langle n | \psi \rangle, \quad A_{mn} = \langle m | \hat{A} | n \rangle, \quad \psi(\mathbf{r}) = \langle \mathbf{r} | \psi \rangle.$$

In position space, $\hat{\mathbf{p}} = -i\hbar\nabla$ yields $[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}\hat{I}$ by the product rule. The lecture uses the simultaneous unitary map

$$|\psi\rangle \mapsto U|\psi\rangle, \quad \hat{A} \mapsto U\hat{A}U^\dagger, \quad U^\dagger U = UU^\dagger = \hat{I}.$$

Supplement—invariance check: $\langle\psi|U^\dagger(U\hat{A}U^\dagger)U|\psi\rangle = \langle\psi|\hat{A}|\psi\rangle$; eigenvalues are unchanged as well. Do not transform the state alone and then claim invariance for a fixed observable. Passive basis conventions may put daggers in the opposite positions: be consistent.

1.8 Born rule, measurement, and Hilbert space

Source PDF pp. 9–10.

Evolution is not the measurement rule. For a normalized wave function, $\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2$ is a probability density, not a probability at a point. In an orthonormal eigenbasis of an observable,

$$\hat{A}|a_n\rangle = a_n|a_n\rangle, \quad |\psi\rangle = \sum_n c_n|a_n\rangle, \quad \boxed{P(a_n) = |c_n|^2 = |\langle a_n|\psi\rangle|^2}.$$

The displayed probability is for a nondegenerate eigenvalue (or a specified resolved basis state). Amplitudes interfere; only their squared moduli give probabilities. For a statistical ensemble of normalized states,

$$\boxed{\hat{\rho} = \sum_i p_i |\phi_i\rangle\langle\phi_i|}, \quad p_i \geq 0, \quad \sum_i p_i = 1, \quad \boxed{\langle A \rangle = \text{Tr}(\hat{\rho}\hat{A})}, \quad S = -k_B \text{Tr}(\hat{\rho} \ln \hat{\rho}).$$

For a pure state, $\hat{\rho} = |\psi\rangle\langle\psi|$ and $S = 0$. The ensemble vectors need not be mutually orthogonal.

Supplement—normalization and projectors. $P(\mathbf{r} \in \Omega) = \int_\Omega |\psi|^2 d^3r$ and $\int |\psi|^2 d^3r = 1$. For the full eigenspace of a possibly degenerate eigenvalue a , let P_a be its orthogonal projector. Ideal selective projective measurement has

$$\text{Pr}(a) = \text{Tr}(\hat{\rho}P_a), \quad \hat{\rho} \mapsto \frac{P_a \hat{\rho} P_a}{\text{Tr}(\hat{\rho}P_a)} \quad \text{if } \text{Pr}(a) > 0.$$

Thus degeneracy requires summing over the eigenspace, not choosing one arbitrary basis vector. An outcome with zero probability has no defined conditioned state. The projection rule is a measurement update, not ordinary closed-system Schrödinger evolution.

Supplement—pure versus mixed. Density operators are positive, Hermitian, and have trace one. If their eigenvalues are r_j , then $S = -k_B \sum_j r_j \ln r_j$, with $0 \ln 0 = 0$; $\text{Tr} \hat{\rho}^2 = 1$ for a pure state and is less than one for a mixed state. In an orthonormal two-state basis, the coherent state $(|0\rangle + |1\rangle)/\sqrt{2}$ and the equal incoherent mixture have matrices

$$\hat{\rho}_{\text{pure}} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad S = 0; \quad \hat{\rho}_{\text{mix}} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S = k_B \ln 2.$$

They have identical probabilities in this basis but different coherence and predictions in other bases. Ensemble weights p_i are not generally eigenvalues of $\hat{\rho}$ when the $|\phi_i\rangle$ are nonorthogonal.

Foundational limit (lecture): the projection postulate is operationally useful without deciding whether collapse is a literal physical process. Von Neumann's early hidden-variable argument used restrictive assumptions; Bell inequalities constrain local hidden-variable theories, not every conceivable additional-variable theory.

1.9 The Dirac equation: quantum mechanics meets relativity

Source PDF pp. 10–11.

Relativistic first-order time evolution requires a multicomponent spinor:

$$\boxed{i\hbar\partial_t\psi = (c\boldsymbol{\alpha} \cdot \hat{\mathbf{p}} + \beta mc^2)\psi}, \quad (i\hbar\gamma^\mu\partial_\mu - mc)\psi = 0, \quad \{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\hat{I}.$$

The Clifford algebra ensures $E^2 = p^2c^2 + m^2c^4$. The lecture's key consequences are spinor structure, electron spin, hydrogen fine structure, and a negative-energy branch whose interpretation led to antiparticles. Relativistic structure is not obtained simply by changing E in the nonrelativistic Schrödinger equation.

Supplement—explicit convention and reconstruction. The lecture leaves the metric implicit; choose $g = \text{diag}(1, -1, -1, -1)$, $x^0 = ct$, $\beta = \gamma^0$, $\alpha^i = \gamma^0\gamma^i$. Then $\{\alpha^i, \alpha^j\} = 2\delta_{ij}\hat{I}$, $\{\alpha^i, \beta\} = 0$, and $\beta^2 = \hat{I}$ give, for the free Hamiltonian,

$$\hat{H}_D^2 = c^2\hat{\mathbf{p}}^2 + m^2c^4, \quad E = \pm\sqrt{p^2c^2 + m^2c^4}.$$

Cross terms vanish by anticommutation, not by setting the matrices to zero. Here $\partial_0 = c^{-1}\partial_t$, explaining the mc in the covariant equation versus mc^2 in the Hamiltonian. The simple squared free Hamiltonian should not be applied unchanged with interactions.

1.10 The Dirac–Feynman path integral: a third formulation

Source PDF pp. 11.

Core meaning. The propagator is the position-space matrix element of the same unitary evolution operator used in the Schrödinger formulation:

$$K(x_b, t_b; x_a, t_a) = \langle x_b | e^{-i\hat{H}(t_b-t_a)/\hbar} | x_a \rangle, \quad \psi(x_b, t_b) = \int dx_a K(x_b, t_b; x_a, t_a) \psi(x_a, t_a).$$

It is a transition *amplitude density*; one must add amplitudes over indistinguishable alternatives before taking an absolute square. For fixed endpoints, the lecture rewrites this same object schematically as

$$K(x_b, t_b; x_a, t_a) = \int_{x(t_a)=x_a}^{x(t_b)=x_b} \mathcal{D}x(t) e^{iS[x]/\hbar}, \quad S[x] = \int_{t_a}^{t_b} L(x, \dot{x}, t) dt.$$

Here a “path” is a complete intermediate history $x(t)$ satisfying the two endpoint conditions. The symbol $\mathcal{D}x$ is not an ordinary one-variable integral: it is defined by time slicing. Every path contributes a unit-modulus phase, not a classical probability; different paths interfere.

Supplement—derive the path sum from the operator propagator. Take $T = t_b - t_a = N\epsilon$, insert $N - 1$ position resolutions of identity, and use the composition law:

$$K = \lim_{N \rightarrow \infty} \int \prod_{j=1}^{N-1} dx_j \prod_{j=0}^{N-1} \langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle, \\ x_0 = x_a, \quad x_N = x_b.$$

For $\hat{H} = \hat{p}^2/(2m) + V(\hat{x})$, first-order Trotter splitting and one momentum resolution give the short-time kernel

$$\langle x_{j+1} | e^{-i\epsilon \hat{H}/\hbar} | x_j \rangle = \int \frac{dp_j}{2\pi\hbar} e^{\frac{i}{\hbar}[p_j(x_{j+1}-x_j) - \epsilon(p_j^2/2m + V(x_j))]} + O(\epsilon^2) \\ = \left(\frac{m}{2\pi i \hbar \epsilon}\right)^{1/2} e^{\frac{i\epsilon}{\hbar}[\frac{m}{2}((x_{j+1}-x_j)/\epsilon)^2 - V(x_j)]} + O(\epsilon^2).$$

Consequently,

$$K = \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \epsilon}\right)^{N/2} \int \prod_{j=1}^{N-1} dx_j e^{iS_N/\hbar}, \quad S_N = \sum_{j=0}^{N-1} \epsilon \left[\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\epsilon} \right)^2 - V(x_j) \right].$$

Thus “sum over all paths” means an ordinary multiple integral over every intermediate position, followed by $N \rightarrow \infty$. The prefactor is part of $\mathcal{D}x$; omitting it loses normalization, dimensions, and the $T \rightarrow 0$ limit $K \rightarrow \delta(x_b - x_a)$. The displayed discretization is for the standard scalar-potential Hamiltonian; velocity-dependent interactions require care over operator ordering and the time-slicing prescription.

Why the classical path emerges without deleting quantum paths. Actions differing by much more than $\hbar$ typically produce rapidly oscillating relative phases and cancellation. For a variation $x \mapsto x + \eta$ with fixed endpoints, $\eta(t_a) = \eta(t_b) = 0$, integration by parts gives

$$\delta S = \int_{t_a}^{t_b} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \eta dt.$$

Hence $\delta S = 0$ gives the Euler–Lagrange equation. Near such a path, $S[x_{cl} + \eta] = S[x_{cl}] + \frac{1}{2}\delta^2 S + \dots$: the linear phase variation vanishes, so nearby contributions remain coherent to leading order. Nonstationary neighborhoods cancel more strongly. The criterion concerns action *variations* relative to $\hbar$; adding a large constant to all actions does not create a classical limit. Stationary does not mean a unique path or an absolute minimum. The functional integral still includes every path; classical mechanics appears by stationary-phase interference, not because nonclassical paths acquire zero probability.

Anchor example—free particle. For $V = 0$ the Gaussian time-sliced integrals give exactly

$$K_0(x_b, t_b; x_a, t_a) = \sqrt{\frac{m}{2\pi i \hbar T}} \exp \left[\frac{i}{\hbar} \frac{m(x_b - x_a)^2}{2T} \right], \quad T > 0.$$

The exponent is the classical action of the straight path, while the prefactor is the fluctuation contribution from paths around it. Checks: $[K_0] = L^{-1}$, $K_0 \rightarrow \delta(x_b - x_a)$ distributionally as $T \rightarrow 0^+$, and convolution of two free kernels gives the kernel for the summed time interval.

Self-check: reconstruct before consulting the key

1. **1.1, 1.11–1.12:** Why are old orbit quantization rules insufficient? Name the distinct roles of dynamics, measurement, and representation.
2. **1.2:** Starting with $Z = (1 - e^{-x})^{-1}$, obtain the mean mode energy and both limits of u_ν . Which extrapolation causes the ultraviolet catastrophe?
3. **1.3–1.4:** Recover the energy and momentum operators from the plane wave. For hydrogen emission, which is positive: $E_{n_i} - E_{n_f}$ or its negative? What does $h\nu < W$ imply for photoemission?
4. **1.5:** For $\hat{H} = \hat{p}^2/(2m) + V(\hat{q})$, derive both Heisenberg canonical equations and state the extra condition for $[\hat{H}, \hat{A}] = 0$ to imply that $\hat{A}$ is conserved.
5. **1.5.1:** What does $\Delta q \Delta p \geq \hbar/2$ describe? Does $\langle [\hat{A}, \hat{B}] \rangle = 0$ imply $[\hat{A}, \hat{B}] = 0$?
6. **1.6:** When may you separate off $e^{-iEt/\hbar}$? Why is the operator-substitution argument not a classical derivation of quantum mechanics?
7. **1.7:** Show an expectation value is invariant under the lecture's simultaneous unitary map. Is a wave function a different physical theory from a matrix?
8. **1.8:** How do you compute and condition on a degenerate measurement outcome? Can a pure superposition and a mixture agree in one measurement basis?
9. **1.9:** Which anticommutation relations remove the cross terms in $\hat{H}_D^2$? Why does the covariant equation contain mc rather than mc^2 ?
10. **1.10:** Starting from $\langle x_b | e^{-i\hat{H}T/\hbar} | x_a \rangle$, where do the integrations over intermediate positions enter? Why are path *amplitudes*, rather than path probabilities, added? What endpoint condition is used in varying the action, and what must the kernel approach as $T \rightarrow 0^+$?

Compact answers

1. They mix classical trajectories with imposed quantum rules and lack general dynamics. Schrödinger gives evolution; Born gives probabilities; unitary representation changes preserve physical predictions.
2. $\bar{E} = -h\nu \partial_x \ln Z = h\nu/(e^x - 1)$. For small x , $e^x - 1 \simeq x$; for large x , $e^x - 1 \simeq e^x$. Extending the resulting $u_\nu \propto \nu^2 T$ to infinite frequency diverges.
3. $i\hbar \partial_t \psi = E\psi$ and $-i\hbar \nabla \psi = \mathbf{p}\psi$. Emission: $n_i > n_f$, $E_{n_i} - E_{n_f} > 0$. Below threshold: no photoemission.
4. $[\hat{p}^2, \hat{q}] = -2i\hbar \hat{p}$ and $[V, \hat{p}] = i\hbar V'$ give $\dot{\hat{q}} = \hat{p}/m$, $\dot{\hat{p}} = -V'$. Also require no explicit time dependence.
5. Intrinsic ensemble spreads in the specified state, not apparatus error. No: a particular expectation can vanish for a nonzero operator.
6. For an energy eigenstate of time-independent $\hat{H}$. Substitution motivates a new dynamical law; classical mechanics does not supply that postulate.
7. Cancel $U^\dagger U$ in the transformed expectation. No: wave functions and matrices are representations of the same abstract theory.
8. $\text{Pr}(a) = \text{Tr}(\hat{\rho} P_a)$ and $\hat{\rho}_a = P_a \hat{\rho} P_a / \text{Pr}(a)$. Yes; coherences distinguish them in other bases.
9. $\{\alpha^i, \alpha^j\} = 2\delta_{ij} \hat{I}$, $\{\alpha^i, \beta\} = 0$, $\beta^2 = \hat{I}$. With $x^0 = ct$, multiplying the covariant equation by $c\gamma^0$ gives Hamiltonian units.
10. Insert $N - 1$ identities $\int dx_j |x_j\rangle \langle x_j|$ between short-time evolution factors. Alternatives are indistinguishable, so complex amplitudes add before $|\cdot|^2$; their relative phases cause interference. Endpoint variations vanish, $\eta(t_a) = \eta(t_b) = 0$, and $K \rightarrow \delta(x_b - x_a)$ distributionally.

Chapter 2: Experimental Clues and Fundamental Concepts

Exam minimum, source PDF pp. 15–40. Keep $\hbar = h/(2\pi)$ explicit; SI units unless stated otherwise. $h, \hbar$ have units of action, spin has units of angular momentum, and Pauli matrices are dimensionless. Hats denote operators when distinction from their matrix representations matters. Discrete-spectrum statements below assume a complete orthonormal eigenbasis; differential operators require appropriate domains and boundary conditions.

2.1 From experimental facts to the quantum description

Source PDF pp. 15–16.

2.1.1 Experimental map. Interference/diffraction of light *and matter* demand amplitudes and phases; black-body radiation, photoemission and Compton scattering demand quanta; spectra and Franck–Hertz demand discrete internal energies; Stern–Gerlach (SG) demands discrete spin-component outcomes, preparation, and incompatible measurements. These are constraints on a theory, not a derivation of the state space or Born rule. Wave/particle behavior refers to experimental outcomes, not a switch between two classical objects.

2.2 Several experiments that reveal quantum behavior

Source PDF pp. 16–19.

2.2.1 Black-body radiation. An ideal absorber/emitter in thermal equilibrium has spectral radiance

$$B_\nu(T) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/(k_B T)} - 1}, \quad E_n = n h \nu \quad (n = 0, 1, \dots).$$

Here B_ν is per unit frequency, with units $\text{W m}^{-2} \text{sr}^{-1} \text{Hz}^{-1}$; do not use this as a per-wavelength spectrum without a Jacobian. Quantized *exchange* in units $h\nu$ defeats classical equipartition at high frequency. This historical E_n convention is not the later harmonic oscillator's full $(n + \frac{1}{2})\hbar\omega$ spectrum. *Use:* form $u = h\nu/(k_B T)$; for $u \ll 1$, $B_\nu \simeq 2\nu^2 k_B T / c^2$, while $u \gg 1$ yields exponential suppression.

2.2.2 Photoelectric effect. With positive elementary charge e and work function Φ ,

$$E_\gamma = h\nu, \quad K_{\max} = h\nu - \Phi = eV_s, \quad \nu_0 = \Phi/h.$$

V_s denotes the positive *magnitude* of the stopping voltage. In the single-photon regime no emission occurs for $\nu < \nu_0$. Above threshold, frequency controls maximum electron energy; intensity primarily controls photon arrival rate and hence current, not $K_{\max}$. *Use:* the slope of V_s versus ν is h/e and the intercept is $-\Phi/e$. Use joules throughout, or convert consistently to electron-volts.

2.2.3 Compton scattering. For a photon scattering through angle θ from an initially stationary, effectively free electron,

$$E_\gamma = pc = h\nu, \quad p_\gamma = h/\lambda, \quad \boxed{\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)}.$$

Energy *and vector momentum* conservation, with relativistic electron energy, give this result. $\lambda_C = h/(m_e c) \simeq 2.426 \times 10^{-12} \text{ m}$. The shift is zero forward, λ_C at 90° , and $2\lambda_C$ backward; λ' is the outgoing photon wavelength. Do not substitute the electron recoil angle or assume the same formula for a tightly bound/moving electron.

2.2.4 Franck–Hertz. Accelerated electrons in a dilute vapor lose energy in inelastic atomic excitation only when they can supply an allowed spacing

$$\Delta E = E_n - E_m > 0, \quad \Delta E \simeq e \Delta V.$$

Successive current drops/recoveries indicate repeated excitations. Use the *spacing* of voltage features to infer level gaps; voltage offsets and collection geometry can move absolute thresholds. Low-energy predominantly elastic collisions do not produce these discrete losses.

2.2.5 What is established? Discrete exchanges, wave-like and particle-like phenomena, discrete atomic internal energies, and h as the wave/mechanical conversion scale. These experiments alone do not prescribe quantum states, observable operators, or outcome probabilities.

2.3 The Stern–Gerlach experiment

Source PDF pp. 19–24.

2.3.1 Two-state system and deflection. Neutral Ag has configuration $[\text{Kr}] 4d^{10}5s^1$: closed shells give no net electronic angular momentum, and the outer electron has $l = 0$, $s = 1/2$. In the course’s effective two-state approximation,

$$H_{\text{int}} = -\boldsymbol{\mu} \cdot \mathbf{B}, \quad \mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}), \quad F_z \simeq \mu_z \partial_z B_z.$$

A uniform field can produce torque but not this spatial splitting. For magnet length L , atom mass M , approximately constant longitudinal speed v , weak deflection and approximately constant gradient,

$$\Delta t \simeq L/v, \quad \Delta v_z \simeq F_z L/(Mv), \quad \theta_z \simeq \frac{\mu_z L \partial_z B_z}{Mv^2} = \frac{\mu_z L \partial_z B_z}{2E_k}.$$

Here $E_k = Mv^2/2$ is *translational*, not spin energy. A downstream drift distance D gives $z \simeq F_z L(D + L/2)/(Mv^2)$ if initial transverse velocity is zero. Thermal $E_k \sim 3k_B T/2$ is only a scale estimate: an effusive beam has a velocity distribution. Classically $\mu_z = \mu \cos \theta$ would span a continuum; experiment gives two branches identified with $S_z = \pm \hbar/2$ in the modern theory. Branch displacement measures μ_z ; an electron’s magnetic moment is opposite its spin, so do not label the upper spot “spin up” without the sign of the coupling and gradient. SG is not a precision determination of $\hbar$.

2.3.2 Sequential analyzers. Selecting the $z+$ branch prepares $|z+\rangle$. Then

$$P(z+|z+) = 1, \quad P(x\pm|z+) = \frac{1}{2}, \quad P(z\pm|x+) = \frac{1}{2}.$$

For $z+ \rightarrow x+ \rightarrow z-$ the final conditional probability given the selected $x+$ beam is $1/2$, but the joint transmission probability relative to the initial $z+$ beam is $1/4$. Multiply conditional probabilities for a sequence of *resolved/selected measurements*; add amplitudes for coherent alternatives that have not been distinguished. The intermediate x filter removes certainty of S_z .

2.3.3 Classical polarization analogy. $\mathbf{E}_{\text{physical}} = \text{Re}[(E_x \hat{\mathbf{x}} + E_y \hat{\mathbf{y}})e^{i(kz - \omega t)}]$. Identify $|z+\rangle \leftrightarrow \hat{\mathbf{x}}$ and $|z-\rangle \leftrightarrow \hat{\mathbf{y}}$. Equal real superpositions represent $\pm 45^\circ$ linear polarization; $(1, \pm i)/\sqrt{2}$ represent circular polarization. Circular handedness labels depend on viewing/time conventions. This is a classical two-component complex-vector analogy, not an appeal to photons. The source’s $|x-\rangle = (-|z+\rangle + |z-\rangle)/\sqrt{2}$ here differs by overall minus sign from its later convention; both represent the same ray.

2.4 State vectors and Dirac notation

Source PDF pp. 24–26.

2.4.1 Kets, bras, inner products. A pure state is a ray: $|\alpha\rangle \sim c|\alpha\rangle$, $c \neq 0$; choose $\langle\alpha|\alpha\rangle = 1$. Only a global phase remains between normalized representatives. Conjugate *all* coefficients when forming a bra:

$$|\psi\rangle = \sum_n c_n |n\rangle \Rightarrow \langle\psi| = \sum_n c_n^* \langle n|, \quad \langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*, \quad \langle\alpha|\alpha\rangle \geq 0.$$

Zero inner product means orthogonality; zero norm means the zero vector, not a physical state. Inner products are linear in the ket, conjugate-linear in the bra.

2.4.2 Operators and adjoints. $A(c|\alpha\rangle + d|\beta\rangle) = cA|\alpha\rangle + dA|\beta\rangle$; $\langle\beta|A|\alpha\rangle = \langle\alpha|A^\dagger|\beta\rangle^*$; $(AB)^\dagger = B^\dagger A^\dagger$. Observables are Hermitian ($A^\dagger = A$; self-adjointness matters for unbounded operators). An outer product is an operator: $(|\alpha\rangle\langle\beta|)|\gamma\rangle = |\alpha\rangle\langle\beta|\gamma\rangle$, unlike the scalar inner product.

2.5 Observables, eigenstates, and matrix representations

Source PDF pp. 26–28.

2.5.1 Hermitian eigenproblem. $A|a_n\rangle = a_n|a_n\rangle$. Taking the expectation and its conjugate proves $a_n = a_n^*$. For two eigenvectors,

$$(a_n - a_m)\langle a_m|a_n\rangle = 0.$$

Distinct eigenvalues force orthogonality; equal eigenvalues do not. Choose an orthonormal basis inside each degenerate eigenspace. With r indexing that basis,

$$\begin{aligned} I &= \sum_{a,r} |a,r\rangle\langle a,r|, & P_a &= \sum_r |a,r\rangle\langle a,r|, \\ A &= \sum_a a P_a, & P_a P_b &= \delta_{ab} P_a, \quad P_a^\dagger = P_a. \end{aligned}$$

For nondegenerate levels, $P_n = |a_n\rangle\langle a_n|$. Inserting completeness is the standard way to derive expansions, spectral decompositions, and products:

$$|\psi\rangle = \sum_n |n\rangle\langle n|\psi\rangle = \sum_n c_n |n\rangle, \quad c_n = \langle n|\psi\rangle, \quad \sum_n |c_n|^2 = 1.$$

Pitfall: “distinct eigenvectors are orthogonal” is false without distinct eigenvalues or an explicit orthonormal choice.

2.5.2 Matrix representation. Fix the basis order before doing any algebra. State columns contain c_n , bras are conjugate-transposed rows, and

$$A_{mn} = \langle m|A|n\rangle, \quad (A\psi)_m = \sum_n A_{mn} c_n, \quad (AB)_{mn} = \sum_k A_{mk} B_{kn}, \quad (A^\dagger)_{mn} = A_{nm}^*.$$

Column n is the expansion of $A|n\rangle$. Reconstruct an operator from a matrix by $A = \sum_{mn} A_{mn} |m\rangle\langle n|$. Matrix mechanics is the same operator theory in coordinates, not another physical theory.

2.6 Measurement and the Born rule

Source PDF pp. 28–29.

2.6.1 Ideal projective measurement. For a normalized premeasurement pure state,

$$p(a) = \langle \psi | P_a | \psi \rangle = \|P_a |\psi\rangle\|^2, \quad |\psi_a\rangle = \frac{P_a |\psi\rangle}{\sqrt{p(a)}} \quad [p(a) > 0].$$

For a nondegenerate level, $p(a_n) = |\langle a_n|\psi\rangle|^2$ and the final ray is $|a_n\rangle$. For degeneracy, project onto the *whole* eigenspace; a measurement of A alone need not select a particular basis vector within it. This is the ideal Lüders update, not a claim about every real measuring device. A zero-probability outcome has no defined conditional state.

$$\sum_a p(a) = 1, \quad \langle A \rangle = \langle \psi | A | \psi \rangle = \sum_a a p(a), \quad \langle A^2 \rangle = \sum_a a^2 p(a).$$

For an unnormalized input, divide expectation values and probabilities by $\langle \psi | \psi \rangle$. An expectation is an ensemble average and need not be an allowed single-shot eigenvalue.

2.6.2 Selective preparation and phase. A selected SG output prepares an eigenstate. If $|\psi\rangle = c_+|+\rangle + c_-|-\rangle$, $c_\pm$ are amplitudes, not probabilities. Measuring S_z sees only their magnitudes; other bases reveal their relative phase. For a subsequent projector Q_b ,

$$p(a \text{ then } b) = \|Q_b P_a |\psi\rangle\|^2, \quad p(b|a) = \frac{\|Q_b P_a |\psi\rangle\|^2}{p(a)}.$$

Recipe: normalize input $\rightarrow$ identify the actual measured eigenspaces $\rightarrow$ project $\rightarrow$ compute norm squared $\rightarrow$ renormalize before the next conditional prediction. A selective measurement is not a passive relabeling of coordinates.

2.7 Spin-1/2 and the Pauli matrices

Source PDF pp. 29–31.

2.7.1 S_z basis and ladders. Throughout use the ordered basis $(|+\rangle, |-\rangle) = (|z+\rangle, |z-\rangle)$:

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad S_z = \frac{\hbar}{2}(|+\rangle\langle+| - |-\rangle\langle-|), \quad S_+ = \hbar|+\rangle\langle-|, \quad S_- = \hbar|-\rangle\langle+|.$$

$S_+|-\rangle = \hbar|+\rangle$, $S_-|+\rangle = \hbar|-\rangle$; both annihilate the other basis ket. Ladder operators are not Hermitian observables; $S_+^\dagger = S_-$.

2.7.2 Other axes and conventions.

$$|x\pm\rangle = \frac{|+\rangle \pm |-\rangle}{\sqrt{2}}, \quad |y\pm\rangle = \frac{|+\rangle \pm i|-\rangle}{\sqrt{2}}, \quad S_i = \frac{\hbar}{2}\sigma_i,$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenvalues of σ_i are ± 1 , whereas those of S_i are $\pm\hbar/2$. Equal z -probabilities alone do not fix the relative phase defining an x axis; orthogonality plus a chosen axis/phase convention gives these expressions.

2.7.3 Spin algebra. Summation over repeated k is understood, with $\epsilon_{xyz} = +1$ and antisymmetry:

$$\begin{aligned} \sigma_i \sigma_j &= \delta_{ij} I + i\epsilon_{ijk} \sigma_k, & [S_i, S_j] &= i\hbar \epsilon_{ijk} S_k, \\ \{S_i, S_j\} &= \frac{\hbar^2}{2} \delta_{ij} I, & S_\pm &= S_x \pm iS_y, \\ [S_z, S_\pm] &= \pm \hbar S_\pm, & [S_-, S_+] &= -2\hbar S_z, \\ S_i^2 &= \frac{\hbar^2}{4} I, & S^2 &= \frac{3}{4}\hbar^2 I = s(s+1)\hbar^2 I \quad (s = \tfrac{1}{2}). \end{aligned}$$

Useful reduction: $(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b})I + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$. “Fermion/boson” concerns particle statistics, not whether an operator has a commutator or anticommutator.

Fast spin probabilities. For normalized $(c_+, c_-)^T$,

$$\begin{aligned} P(x\pm) &= \tfrac{1}{2}|c_+ \pm c_-|^2, & P(y\pm) &= \tfrac{1}{2}|c_+ \mp ic_-|^2, \\ \langle S_x \rangle &= \hbar \operatorname{Re}(c_+^* c_-), & \langle S_y \rangle &= \hbar \operatorname{Im}(c_+^* c_-), \\ \langle S_z \rangle &= \tfrac{\hbar}{2}(|c_+|^2 - |c_-|^2), & (\Delta S_i)^2 &= \hbar^2/4 - \langle S_i \rangle^2. \end{aligned}$$

The opposite sign in the y amplitude comes from conjugating the eigenket.

2.8 Compatible and incompatible observables

Source PDF pp. 31.

In the discrete finite-dimensional setting, Hermitian A, B admit a complete common orthonormal eigenbasis iff $[A, B] = 0$. In an A eigenbasis,

$$(a_m - a_n)\langle a_m | B | a_n \rangle = 0.$$

If A is nondegenerate this forces B diagonal. If A is degenerate it forces B to be block diagonal; diagonalize B *within each equal- a block*. A common eigenvector need not imply commuting operators; noncommuting operators may have isolated common eigenvectors (e.g. all L_i annihilate an $l=0$ state). Spin-1/2 S_x, S_z have no common eigenvector and $[S_x, S_z] = -i\hbar S_y$. For unbounded operators, formal commutators alone require domain/spectral care.

2.9 Change of basis and unitary transformations

Source PDF pp. 32–33.

2.9.1 Same physical state, new coordinates. In the conventions of 2.7,

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad c_x = U^\dagger c_z, \quad A_x = U^\dagger A_z U.$$

The columns of U are $|x+\rangle, |x-\rangle$ expressed in the old z basis. Here $U = U^\dagger$, accidentally making forward and inverse look identical. Explicitly,

$$S_x^{(x)} = \frac{\hbar}{2}\sigma_z, \quad S_y^{(x)} = -\frac{\hbar}{2}\sigma_y, \quad S_z^{(x)} = \frac{\hbar}{2}\sigma_x.$$

The minus sign for $S_y^{(x)}$ is required by the chosen phases. Rephasing a basis vector changes matrix entries but not observables or probabilities.

2.9.2 General unitary basis map. For complete orthonormal bases $\{|a_n\rangle\}$ and $\{|b_n\rangle\}$,

$$\hat{U} = \sum_n |b_n\rangle\langle a_n|, \quad \hat{U}|a_n\rangle = |b_n\rangle, \quad U_{mn} = \langle a_m|b_n\rangle, \quad U^\dagger U = U U^\dagger = I.$$

The matrix U contains new basis vectors in old coordinates. Insert completeness to reconstruct the transformation rather than memorizing its direction:

$$(c_b)_n = \langle b_n|\psi\rangle = \sum_m \langle b_n|a_m\rangle (c_a)_m \Rightarrow \boxed{c_b = U^\dagger c_a, \quad A_b = U^\dagger A_a U}.$$

Unitary transformations preserve inner products, norms and probabilities. A *passive* basis change leaves the physical ket fixed and transforms its coordinates by $U^\dagger$. An *active* transformation sends $|\psi\rangle \rightarrow \hat{U}|\psi\rangle$ at fixed basis, hence $c \rightarrow Uc$. Do not mix these operations; similarly $UAU^\dagger$ is the actively conjugated operator, not the passive formula with the above definition of U .

2.9.3 Trace and diagonalization.

$$\text{Tr } A = \sum_n \langle a_n|A|a_n\rangle, \quad \text{Tr}(AB) = \text{Tr}(BA), \quad \text{Tr}(U^\dagger AU) = \text{Tr } A.$$

Cyclicity permits cyclic permutations, not arbitrary reordering. For example $\text{Tr}(ABC) = \text{Tr}(BCA)$, but not generally $\text{Tr}(ACB)$. These statements require finite matrices or suitable trace-class products; do not apply a finite trace argument to $[x, p] = i\hbar I$ on an infinite-dimensional space.

To diagonalize Hermitian B :

1. Solve $\det(B - bI) = 0$ for eigenvalues.
2. Solve $BC^{(l)} = b_l C^{(l)}$ for normalized eigenvectors; orthonormalize inside degenerate subspaces.
3. Put these columns into U : $U_{nl} = C_n^{(l)} = \langle a_n|b_l\rangle$.
4. Check $U^\dagger U = I$ and $U^\dagger BU = \text{diag}(b_l)$; use the same column order for eigenvalues, states and amplitudes.

Similarity preserves eigenvalues, trace and determinant. A basis change cannot alter predicted outcomes.

Mini-check of the recipe. The $z+$ state has old column $(1, 0)^T$ and new column $(1, 1)^T/\sqrt{2}$ in the x basis. It still has definite S_z : applying $S_z^{(x)} = (\hbar/2)\sigma_x$ to that column returns $(\hbar/2)$ times the same column. Its nontrivial coordinates in the x basis do not mean a measurement has occurred.

2.10 Continuous spectra: position and momentum

Source PDF pp. 33–36.

2.10.1 Generalized eigenstates and delta functions.

$$\hat{\xi}|\xi\rangle = \xi|\xi\rangle, \quad \langle\xi|\xi'\rangle = \delta(\xi - \xi'), \quad \int d\xi |\xi\rangle\langle\xi| = I, \quad |\psi\rangle = \int d\xi |\xi\rangle\langle\xi|\psi\rangle.$$

These are distributional eigenstates, not unit-norm Hilbert-space vectors. Integrals below run over the real line unless bounds are given. For smooth test functions,

$$\begin{aligned} \int dx \delta(x-a)f(x) &= f(a), & \int dx \delta'(x-a)f(x) &= -f'(a), \\ \delta(-x) &= \delta(x), & \delta(ax) &= \delta(x)/|a| \quad (a \neq 0), \\ x\delta(x) &= 0, & x\delta'(x) &= -\delta(x), \\ \delta(x^2 - a^2) &= \frac{\delta(x-a) + \delta(x+a)}{2|a|} \quad (a \neq 0), & \delta(x) &= \frac{1}{2\pi} \int dk e^{ikx}. \end{aligned}$$

A useful extension for simple roots x_j is $\delta(g(x)) = \sum_j \delta(x - x_j)/|g'(x_j)|$; it fails at multiple roots without further distributional analysis. $\delta(x)$ has units of inverse length. Do not treat $\delta(0)$ as a finite normalization or square a delta to obtain a probability.

2.10.2 Position representation.

$$\psi(x) = \langle x|\psi\rangle, \quad \int dx |\psi(x)|^2 = 1, \quad P(a < x < b) = \int_a^b dx |\psi(x)|^2, \quad (\hat{x}\psi)(x) = x\psi(x).$$

ψ has units $\text{length}^{-1/2}$ in one dimension. $|\psi(x)|^2$ is a density, not a probability at an exact point. A finite-resolution ideal position measurement uses an interval projector $P_{[a,b]} = \int_a^b dx |x\rangle\langle x|$.

2.10.3 Translation and its generator. Define an *active* translation by $T(a)|x\rangle = |x+a\rangle$. Then $T(0) = I$, $T(a)T(b) = T(a+b)$ and $T(a)^\dagger = T(-a) = T(a)^{-1}$.

$$T(\delta x) = I - iK\delta x + O(\delta x^2), \quad K^\dagger = K, \quad [x, K] = i, \quad p = \hbar K = -i\hbar\partial_x,$$

$$\boxed{T(a) = e^{-iap/\hbar}, \quad (T(a)\psi)(x) = \psi(x-a), \quad [x, p] = i\hbar I.}$$

The minus sign in the translated wave function shifts its peak to the *right* by a . Check $T^\dagger x T = x + aI$, or directly $[x, -i\hbar\partial_x]\psi = i\hbar\psi$. K has units inverse length; p has momentum units. Unitarity/strong continuity underlie the Hermitian generator; differentiability and operator domains are implicit.

2.10.4 Fourier-related wave functions. Choose the source's phase convention:

$$\begin{aligned} \phi(p) &= \langle p|\psi\rangle, & \langle x|p\rangle &= \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}, \\ \psi(x) &= \frac{1}{\sqrt{2\pi\hbar}} \int dp e^{ipx/\hbar} \phi(p), & \phi(p) &= \frac{1}{\sqrt{2\pi\hbar}} \int dx e^{-ipx/\hbar} \psi(x). \end{aligned}$$

Here $\langle p|p'\rangle = \delta(p - p')$, $\int dp |\phi(p)|^2 = 1$, and ϕ has units $\text{momentum}^{-1/2}$. In momentum space p multiplies and $x = i\hbar\partial_p$. Use complex-conjugate Fourier kernels for inverse directions; changing from p to wave number $k = p/\hbar$ changes the normalization and measure. Integration by parts requires vanishing boundary terms on the chosen domain.

2.11 The uncertainty relation

Source PDF pp. 36–38.

2.11.1 Variance and Robertson inequality. Let $\delta A = A - \langle A \rangle I$ and $\Delta A = \sqrt{\langle (\delta A)^2 \rangle}$; the former is an operator, the latter a nonnegative number. For normalized states with the required finite moments/products,

$$\begin{aligned} (\Delta A)^2 &= \langle A^2 \rangle - \langle A \rangle^2, \\ (\Delta A)^2 (\Delta B)^2 &\geq |\langle \delta A \delta B \rangle|^2 = \frac{1}{4} |\langle [A, B] \rangle|^2 + \frac{1}{4} |\langle \{\delta A, \delta B\} \rangle|^2, \\ \boxed{\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|}, \quad &\boxed{\Delta x \Delta p \geq \hbar/2}. \end{aligned}$$

Derivation: Cauchy–Schwarz on $\delta A|\psi\rangle, \delta B|\psi\rangle$; split their product into commutator plus anticommutator. Their expectations are respectively imaginary and real; drop the nonnegative covariance term for Robertson. Saturating Cauchy–Schwarz alone need not saturate Robertson: also require zero symmetrized covariance. For nonzero variances, Robertson equality requires the two fluctuation vectors proportional with a purely imaginary coefficient. A nonzero commutator can have zero expectation in a particular state, giving a zero lower bound without compatibility. These are *state spreads*, not instrumental error or a disturbance formula.

2.11.2 Limiting cases. A momentum eigenstate is a plane wave $e^{ipx/\hbar}/\sqrt{2\pi\hbar}$, fully delocalized; a position eigenstate has momentum amplitude $e^{-ipx_0/\hbar}/\sqrt{2\pi\hbar}$, constant in magnitude. Both are nonnormalizable idealizations: do not substitute “ $0 \times \infty$ ” into the variance inequality. Interpret them as limits of normalized packets. For the source’s real centered Gaussian ($x_0 > 0$ is a *width*, not its center),

$$\psi(x) = \frac{e^{-x^2/(2x_0^2)}}{\pi^{1/4}\sqrt{x_0}}, \quad \phi(p) = \frac{\sqrt{x_0}}{\pi^{1/4}\sqrt{\hbar}} e^{-p^2 x_0^2/(2\hbar^2)}, \quad \Delta x = \frac{x_0}{\sqrt{2}}, \quad \Delta p = \frac{\hbar}{\sqrt{2}x_0}.$$

Square the amplitude *before* identifying a normal-distribution variance. This unchirped Gaussian saturates $\hbar/2$; a Gaussian with a quadratic phase need not saturate this product.

2.12 Worked examples

Source PDF pp. 38–39.

2.12.1 General Hermitian 2×2 matrix. For real $a_0, \mathbf{a}$,

$$X = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma} = \begin{pmatrix} a_0 + a_z & a_x - ia_y \\ a_x + ia_y & a_0 - a_z \end{pmatrix}, \quad \det(X - \lambda I) = (a_0 - \lambda)^2 - |\mathbf{a}|^2.$$

Thus $\lambda_{\pm} = a_0 \pm |\mathbf{a}|$. Read $a_y = -\text{Im } X_{12}$, not the opposite sign. Alternatively $(\mathbf{a} \cdot \boldsymbol{\sigma})^2 = |\mathbf{a}|^2 I$. When $|\mathbf{a}| > 0$, projectors $P_{\pm} = (I \pm \hat{\mathbf{a}} \cdot \boldsymbol{\sigma})/2$ give $X = \lambda_+ P_+ + \lambda_- P_-$. When $\mathbf{a} = 0$, $X = a_0 I$ is fully degenerate and $\hat{\mathbf{a}}$ is undefined; any orthonormal basis works.

2.12.2 Distributional matrix elements: source sign correction. With the stated Fourier convention the correct kernels are

$$\boxed{\langle p|x|p'\rangle = i\hbar\partial_p\delta(p-p') = -i\hbar\partial_{p'}\delta(p-p'), \quad \langle x|p|x'\rangle = -i\hbar\partial_x\delta(x-x')}.$$

The original PDF’s Eqs. (2.172)–(2.173) have both signs reversed. Indeed differentiate $e^{i(p'-p)x/\hbar}$ with respect to p under $\int dx/(2\pi\hbar)$, or integrate the kernel against $\phi(p')$ to obtain $i\hbar\phi'(p)$. Then

$$\langle p|[x, p]|p'\rangle = (p' - p)i\hbar\partial_p\delta(p-p') = i\hbar\delta(p-p').$$

Always specify which variable is differentiated; integration against a test function is the safest sign check.

2.13 Problems: solution roadmaps and final checks

Source PDF pp. 39–40.

1. **Spin spectral decomposition and phases.** Insert $I = |+\rangle\langle+| + |-\rangle\langle-|$ into $S_z I$ and apply each eigenvalue equation to obtain $S_z = (\hbar/2)(|+\rangle\langle+| - |-\rangle\langle-|)$. Equal probabilities and orthogonality allow

$$|x+\rangle = \frac{|+\rangle + e^{i\chi}|-\rangle}{\sqrt{2}}, \quad |x-\rangle = \frac{|+\rangle - e^{i\chi}|-\rangle}{\sqrt{2}},$$

up to independent overall phases. Setting $\chi = 0$ chooses the usual x axis/relative phase of the z basis. *The probabilities and orthogonality alone do not determine χ in a fixed z -basis convention*; the problem’s “up to an overall phase” needs this extra convention. Check zero overlap and unit norm explicitly.

2. **Cauchy–Schwarz and minimum uncertainty.** For $\beta \neq 0$, minimize $\|\alpha + \lambda\beta\|^2$ with

$$\lambda = -\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle} \Rightarrow |\langle\beta|\alpha\rangle|^2 \leq \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle.$$

Equality iff the vectors are linearly dependent (including the trivial zero-vector case). Apply to fluctuation vectors and follow 2.11’s commutator split. For $d > 0$, write $\bar{x} = \langle x \rangle$, $\bar{p} = \langle p \rangle$ in

$$\psi(x) = (2\pi d^2)^{-1/4} \exp\left[\frac{i\bar{p}x}{\hbar} - \frac{(x - \bar{x})^2}{4d^2}\right].$$

$|\psi|^2$ is a normal density with mean $\bar{x}$ and variance d^2 . Differentiate once:

$$(p - \bar{p})\psi = \frac{i\hbar(x - \bar{x})}{2d^2}\psi, \quad \|(p - \bar{p})\psi\|^2 = \frac{\hbar^2}{4d^2}.$$

Hence $\Delta x = d$, $\Delta p = \hbar/(2d)$, product $\hbar/2$. The proportionality is imaginary, so covariance vanishes as required.

3. **Simultaneous diagonalization with degeneracy.** For the given $A = \text{diag}(a, -a, -a)$ and $B = b \oplus b\sigma_y$ in $(|1\rangle, |2\rangle, |3\rangle)$, the common orthonormal basis and eigenvalue pairs are

ket	A	B
$ 1\rangle$	a	b
$(2\rangle + i 3\rangle)/\sqrt{2}$	$-a$	b
$(2\rangle - i 3\rangle)/\sqrt{2}$	$-a$	$-b$

For $b \neq 0$, B has eigenvalue b twice and $-b$ once; for $b = 0$ all three are zero. $[A, B] = 0$ because the only off-diagonal B entries lie within the equal- $-a$ block. For $a \neq 0$, A has a twofold $-a$ eigenspace; if $a = 0$, it is fully degenerate. The displayed basis works even when a or b vanishes. Degeneracy gives freedom to diagonalize the other operator, not an obstruction.

4. **Optional: rotation of a Pauli vector.** Require real unit $\mathbf{n}$. Since $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$,

$$U(\varphi) = \cos(\varphi/2)I - i\sin(\varphi/2)\mathbf{n} \cdot \boldsymbol{\sigma}.$$

Expand $U(\mathbf{a} \cdot \boldsymbol{\sigma})U^\dagger$ using the Pauli product identity; collect coefficients to obtain the active right-handed Rodrigues rotation

$$\mathbf{a}' = \mathbf{a} \cos \varphi + (\mathbf{n} \times \mathbf{a}) \sin \varphi + \mathbf{n}(\mathbf{n} \cdot \mathbf{a})(1 - \cos \varphi).$$

For $\mathbf{n} = \hat{\mathbf{z}}$, $U = \text{diag}(e^{-i\varphi/2}, e^{i\varphi/2})$ and $(a'_x, a'_y, a'_z) = (a_x \cos \varphi - a_y \sin \varphi, a_x \sin \varphi + a_y \cos \varphi, a_z)$. Check $U\sigma_x U^\dagger = \cos \varphi \sigma_x + \sin \varphi \sigma_y$. Reversing the conjugation reverses the angle. $U(2\pi) = -I$ leaves the physical ray and the rotated Pauli vector unchanged.

Self-checks (answers below). (i) Can increasing intensity overcome the single-photon threshold? (ii) What fraction of a prepared $z+$ beam survives $x+$ then $z-$? (iii) Is $(1, i)^T/\sqrt{2}$ an $x+$ state? (iv) Does $[A, B] \neq 0$ forbid every common eigenket? (v) What happens to $\psi(x)$ under $T(a)$? (vi) Is the source’s Gaussian width x_0 its standard deviation? (vii) Which kernel sign produces $x = i\hbar\partial_p$?

Self-check answers. (i) No; $\hbar\nu \geq \Phi$ is required in that regime. (ii) 1/4 jointly, 1/2 conditional on passing $x+$. (iii) No: it is $y+$, with $P(x\pm) = 1/2$. (iv) No; it forbids a complete common eigenbasis in the stated setting. (v) $\psi(x - a)$, a rightward shift by a . (vi) No: $\Delta x = x_0/\sqrt{2}$. (vii) $+i\hbar\partial_p\delta(p - p')$, equivalently $-i\hbar\partial_{p'}\delta(p - p')$.

Chapter 3: Quantum Dynamics

Scope. Original extract pp. 41–57 covers Sections 3.1–3.4; the supplied continuation pp. 58–74 completes Sections 3.4.1–3.9. The continuation is kept in `chapter3_continuation.tex` so its source boundary remains explicit.

3.1 Time evolution and the Schrödinger equation

Source PDF pp. 41–46.

3.1.1 The time-evolution operator

Setting: closed-system, continuous unitary dynamics; time is a parameter, not a position-like observable. With a common Hamiltonian governing every initial state,

$$\begin{aligned} |\psi(t)\rangle &= U(t, t_0)|\psi(t_0)\rangle, & U^\dagger U &= U U^\dagger = 1, \\ U(t_0, t_0) &= 1, & U(t_2, t_0) &= U(t_2, t_1)U(t_1, t_0), & U^{-1}(t, t_0) &= U(t_0, t) = U^\dagger(t, t_0). \end{aligned}$$

Derivation backbone: continuity gives $U(t + \delta t, t) = 1 - i\Omega(t)\delta t + O(\delta t^2)$. Then $U^\dagger U = 1 + i(\Omega^\dagger - \Omega)\delta t + O(\delta t^2)$ implies $\Omega^\dagger = \Omega$. Identifying the energy generator as $H = \hbar\Omega$ gives

$$U(t + \delta t, t) = 1 - \frac{i}{\hbar}H(t)\delta t + O(\delta t^2).$$

Energy generates time translations, just as momentum generates spatial translations: $T(\delta \mathbf{r}) = 1 - i\delta \mathbf{r} \cdot \mathbf{p}/\hbar + O(\delta r^2)$. Unitarity establishes a Hermitian generator; its identification with the physical energy is additional physical input.

3.1.2 Derivation of the Schrödinger equation

Insert the infinitesimal operator into the composition law, subtract $U(t, t_0)$ and divide by δt :

$$i\hbar\partial_t U(t, t_0) = H(t)U(t, t_0), \quad i\hbar\partial_t |\psi(t)\rangle = H(t)|\psi(t)\rangle.$$

For a local scalar potential and no vector potential, $H = \mathbf{p}^2/(2m) + V(\mathbf{r}, t)$, $\mathbf{p} = -i\hbar\nabla$, so

$$i\hbar\partial_t \psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t) \right] \psi(\mathbf{r}, t).$$

First order in time: one initial state, not an independent initial time derivative, determines the evolution (with the Hamiltonian and spatial domain/boundary conditions specified).

3.1.3 Solutions for the time-evolution operator

Time-independent H : $U = e^{-iH(t-t_0)/\hbar}$; the exponential is an operator power series, not division by H .

Commuting time-dependent H : if $[H(t_1), H(t_2)] = 0$ for every pair of times,

$$U(t, t_0) = \exp \left[-\frac{i}{\hbar} \int_{t_0}^t H(s) ds \right].$$

General noncommuting case (write the following for $t \geq t_0$): integrate the evolution equation and iterate,

$$\begin{aligned} U(t, t_0) &= 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1)U(t_1, t_0) \\ &= 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 H(t_1) + \left(-\frac{i}{\hbar} \right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 H(t_1)H(t_2) + \cdots \\ &= \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t H(s) ds \right]. \end{aligned}$$

$\mathcal{T}$ puts later-time factors on the left. The ordered triangular integral has no extra $1/2!$. Equivalently, integrate the time-ordered product over the full square and include a factor $1/2!$. A magnetic field whose direction changes can give noncommuting spin Hamiltonians. **Pitfall:** a Hermitian $H(t)$ does not imply that Hamiltonians at different times commute; dropping $\mathcal{T}$ is generally wrong.

3.1.4 Energy eigenstates and stationary states

For time-independent H , use a complete orthonormal energy basis, with extra degeneracy labels when needed (and integrals for continuous spectra):

$$H|n\rangle = E_n|n\rangle, \quad \sum_n |n\rangle\langle n| = 1,$$

$$U(t, t_0) = \sum_n e^{-iE_n(t-t_0)/\hbar} |n\rangle\langle n|, \quad |\psi(t)\rangle = \sum_n c_n e^{-iE_n(t-t_0)/\hbar} |n\rangle.$$

The populations $|c_n|^2$ stay fixed; relative phases evolve. To locate measurable oscillations, expand a time-independent observable:

$$\langle A \rangle_t = \sum_{m,n} c_m^* c_n A_{mn} e^{i(E_m - E_n)(t-t_0)/\hbar}, \quad A_{mn} = \langle m|A|n\rangle.$$

Thus energy differences determine frequencies, while coherences and matrix elements determine which frequencies appear. A superposition need not make every observable oscillate.

Stationarity: an energy eigenstate acquires only a global phase, so all time-independent observables have time-independent probability distributions. This is not the assertion that the particle is at rest. Any superposition wholly within one degenerate energy eigenspace is also stationary. Explicitly time-dependent observables are excluded from this statement.

Degeneracy and quantum numbers: a complete set of commuting observables satisfies

$$[A_i, A_j] = 0, \quad [A_i, H] = 0, \quad H|E, a, b, \dots\rangle = E|E, a, b, \dots\rangle.$$

Their simultaneous eigenvalues distinguish states with the same energy; enough independent commuting labels are needed for a complete labeling. Conservation also requires no explicit time dependence of these observables.

3.1.5 Example: spin precession

Keep the source's signed convention, rather than inserting an electron-charge sign by memory:

$$H = -\boldsymbol{\mu} \cdot \mathbf{B} = -\gamma \mathbf{S} \cdot \mathbf{B}, \quad \mathbf{B} = B\hat{\mathbf{z}}, \quad \omega = -\gamma B, \quad H = \omega S_z.$$

For spin 1/2, $S_z = \frac{\hbar}{2} \text{diag}(1, -1)$ and $E_{\pm} = \pm \hbar\omega/2$. Therefore

$$c_+|+z\rangle + c_-|-z\rangle \mapsto c_+e^{-i\omega t/2}|+z\rangle + c_-e^{i\omega t/2}|-z\rangle.$$

Starting from $|+x\rangle = (|+z\rangle + |-z\rangle)/\sqrt{2}$, project onto $|\pm x\rangle = (|+z\rangle \pm |-z\rangle)/\sqrt{2}$:

$$\langle +x|\psi(t)\rangle = \cos(\omega t/2), \quad \langle -x|\psi(t)\rangle = -i \sin(\omega t/2),$$

$$P(S_x = +\hbar/2) = \cos^2(\omega t/2), \quad P(S_x = -\hbar/2) = \sin^2(\omega t/2),$$

$$\langle S_x \rangle_t = \frac{\hbar}{2} \cos \omega t, \quad \langle S_y \rangle_t = \frac{\hbar}{2} \sin \omega t, \quad \langle S_z \rangle_t = 0.$$

Mechanism: the two S_z components retain their weights and develop a relative phase. If the initial state is $|+z\rangle$, there is instead only a global phase and no precessing transverse mean. **Sign check:** with $H = \omega S_z$, $\dot{S}_x = (i/\hbar)[\omega S_z, S_x] = -\omega S_y$ and $\dot{S}_y = \omega S_x$, consistent with the displayed trajectory. The sign of γ determines its sense.

3.2 The Schrödinger and Heisenberg pictures

Source PDF pp. 47–50.

3.2.1 Unitary changes of representation

$U^\dagger U = 1$ preserves inner products. The identity $(\langle\alpha|U^\dagger)X(U|\beta\rangle) = \langle\alpha|(U^\dagger XU)|\beta\rangle$ lets us move evolution from states to operators without changing matrix elements. A change of picture changes representation, not experimental predictions.

3.2.2 Two equivalent pictures

Using the same reference time t_0 and propagator $U(t, t_0)$,

$$\begin{aligned} |\psi_S(t)\rangle &= U|\psi_S(t_0)\rangle, & |\psi_H\rangle &= U^\dagger|\psi_S(t)\rangle = |\psi_S(t_0)\rangle, \\ X_H(t) &= U^\dagger X_S(t) U, & \langle\psi_S(t)|X_S(t)|\psi_S(t)\rangle &= \langle\psi_H|X_H(t)|\psi_H\rangle. \end{aligned}$$

Schrödinger operators may have *explicit* time dependence; they are fixed only when that dependence is absent. Transform all operators consistently: $[A_H, B_H] = U^\dagger[A_S, B_S]U$.

3.2.3 The equation of motion in the Heisenberg picture

Differentiate all three factors of $U^\dagger X_S U$, using $\dot{U} = -iH_S U/\hbar$ and $\dot{U}^\dagger = iU^\dagger H_S/\hbar$:

$$\boxed{\frac{dX_H}{dt} = \frac{i}{\hbar}[H_H, X_H] + U^\dagger \frac{\partial X_S}{\partial t} U}, \quad H_H = U^\dagger H_S U.$$

Equivalently, $i\hbar\dot{X}_H = [X_H, H_H] + i\hbar(\partial_t X_S)_H$. For time-independent H , $H_H = H_S$; do not assume this for arbitrary time-dependent noncommuting $H_S(t)$. **Conservation:** $[X_S, H_S] = 0$ together with $\partial_t X_S = 0$ implies $\dot{X}_H = 0$. Commutation alone is insufficient if X_S explicitly changes with time. In particular $d\langle H\rangle/dt = \langle\partial_t H\rangle$.

3.2.4 Constructing the Hamiltonian and the classical correspondence

Model assumptions: for a nonrelativistic particle in a scalar potential, $H = \mathbf{p}^2/(2m) + V(\mathbf{x})$ and $[x_i, p_j] = i\hbar\delta_{ij}$. Promoting classical variables to operators is not a universal quantization algorithm: operator ordering can be ambiguous; spin is an internal degree of freedom; symmetry and gauge invariance constrain interactions; closed-system evolution requires a Hermitian Hamiltonian. For suitably expandable functions,

$$[x_i, F(\mathbf{p})] = i\hbar \frac{\partial F}{\partial p_i}, \quad [p_i, G(\mathbf{x})] = -i\hbar \frac{\partial G}{\partial x_i}.$$

Free particle: $\dot{\mathbf{p}} = 0$, $\dot{\mathbf{x}} = \mathbf{p}/m$ imply

$$\mathbf{x}(t) = \mathbf{x}(0) + \frac{t}{m}\mathbf{p}(0), \quad [x_i(t), x_j(0)] = -\frac{i\hbar t}{m}\delta_{ij}.$$

Equal-time position components commute; different-time positions need not. **Scalar potential and Ehrenfest:**

$$\begin{aligned} \dot{x}_i &= p_i/m, & \dot{p}_i &= -\partial_i V(\mathbf{x}), & m\ddot{x}_i &= -\partial_i V(\mathbf{x}), \\ \boxed{\frac{d\langle x_i \rangle}{dt} &= \frac{\langle p_i \rangle}{m}, & \frac{d\langle p_i \rangle}{dt} &= -\langle \partial_i V(\mathbf{x}) \rangle.} \end{aligned}$$

Classical limit pitfall: generally $\langle\partial_i V(\mathbf{x})\rangle \neq \partial_i V(\langle\mathbf{x}\rangle)$. Equality is exact for potentials at most quadratic, whose force is linear/constant; it is approximate for a narrow packet in a slowly varying force. Exact classical motion of the mean does not imply a classical probability distribution or zero fluctuations.

3.3 The simple harmonic oscillator

Source PDF pp. 50–56.

Why this model: near a smooth stable minimum x_{eq} with $V''(x_{\text{eq}}) > 0$, $V(x) \simeq V(x_{\text{eq}}) + \frac{1}{2}V''(x_{\text{eq}})(x - x_{\text{eq}})^2$. Shift the origin and discard the constant energy offset; $m\omega^2 = V''(x_{\text{eq}})$, $\omega > 0$. The approximation needs sufficiently small excursions; a flat minimum need not be harmonic. Classically $x = A \cos(\omega t + \delta)$, $p = -m\omega A \sin(\omega t + \delta)$ and $E = m\omega^2 A^2/2$.

3.3.1 Ladder operators and the energy spectrum

Start from $H = p^2/(2m) + m\omega^2 x^2/2$, $[x, p] = i\hbar$. Then $[x, H] = i\hbar p/m$, $[p, H] = -i\hbar m\omega^2 x$, so $[x \pm ip/(m\omega), H] = \pm\hbar\omega[x \pm ip/(m\omega)]$. Define

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(x + \frac{ip}{m\omega} \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{ip}{m\omega} \right), \quad [a, a^\dagger] = 1,$$

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger), \quad p = -i\sqrt{\frac{m\hbar\omega}{2}}(a - a^\dagger), \quad \boxed{H = \hbar\omega(N + \frac{1}{2}), \quad N = a^\dagger a.}$$

Spectrum proof: $\langle N \rangle = \|a|\psi\rangle\|^2 \geq 0$ and $[N, a] = -a$, $[N, a^\dagger] = a^\dagger$. Repeated lowering of an eigenstate must terminate before a negative eigenvalue; the terminating state has norm $\|a|0\rangle\|^2 = 0$, hence eigenvalue zero. The original eigenvalue is therefore an integer. Norms fix the ladder coefficients:

$$a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \quad |n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle,$$

$$\boxed{E_n = \hbar\omega(n + \frac{1}{2}), \quad n = 0, 1, \dots, \quad a|0\rangle = 0.}$$

The vacuum is not the zero vector. Zero-point energy $E_0 = \hbar\omega/2$ reflects the impossibility of simultaneously sharp $x = p = 0$.

3.3.2 Matrix elements of position and momentum

$$\langle n'|x|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}),$$

$$\langle n'|p|n\rangle = i\sqrt{\frac{m\hbar\omega}{2}}(-\sqrt{n}\delta_{n',n-1} + \sqrt{n+1}\delta_{n',n+1}).$$

In the ordered basis $|0\rangle, |1\rangle, \dots$, $a_{n',n} = \sqrt{n}\delta_{n',n-1}$: its first upper off-diagonal is 1, $\sqrt{2}, \sqrt{3}, \dots$; $a^\dagger$ is its conjugate transpose. Linear x or p couplings connect only $\Delta n = \pm 1$; this is not a selection rule for arbitrary nonlinear couplings.

3.3.3 Wave functions and Hermite polynomials

Set $x_0 = \sqrt{\hbar/(m\omega)}$, $\xi = x/x_0$ (here x_0 is a length, not the equilibrium location). Then $a = (\xi + \partial_\xi)/\sqrt{2}$ and $a\psi_0 = 0$ gives $\psi_0 \propto e^{-\xi^2/2}$; normalize in dx , then apply $(a^\dagger)^n/\sqrt{n!}$:

$$\boxed{\psi_n(x) = \frac{H_n(x/x_0)e^{-x^2/(2x_0^2)}}{\pi^{1/4}\sqrt{2^n n!}x_0}.$$

H_n are the physicists' Hermite polynomials. There are n nodes and parity $(-1)^n$; the Gaussian factor uses the same x_0 for all n . Using ladder algebra, $\langle x \rangle_n = \langle p \rangle_n = 0$ and

$$\langle x^2 \rangle_n = \frac{\hbar}{m\omega}(n + \frac{1}{2}), \quad \langle p^2 \rangle_n = m\hbar\omega(n + \frac{1}{2}), \quad \Delta x \Delta p = \hbar(n + \frac{1}{2}).$$

Only the ground number state saturates $\Delta x \Delta p = \hbar/2$; its rms width is $x_0/\sqrt{2}$, not x_0 .

3.3.4 Time evolution of oscillator observables

Heisenberg operators (reference time zero) satisfy

$$\dot{x} = p/m, \quad \dot{p} = -m\omega^2 x, \quad \dot{a} = -i\omega a, \quad \dot{a}^\dagger = i\omega a^\dagger.$$

Solve either the coupled equations or the ladder equations:

$$\begin{aligned} a(t) &= a(0)e^{-i\omega t}, & a^\dagger(t) &= a^\dagger(0)e^{i\omega t}, \\ \boxed{x(t) &= x(0) \cos \omega t + \frac{p(0)}{m\omega} \sin \omega t}, \\ \boxed{p(t) &= -m\omega x(0) \sin \omega t + p(0) \cos \omega t}. \end{aligned}$$

These are operator identities, valid in every state. They do not say every state has an oscillating mean: $\langle n|x(t)|n \rangle = \langle n|p(t)|n \rangle = 0$.

For a normalized $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$ at $t = 0$, the off-diagonal matrix elements alone survive:

$$\begin{aligned} \langle x(t) \rangle &= \sqrt{\frac{\hbar}{2m\omega}} (c_0^* c_1 e^{-i\omega t} + c_1^* c_0 e^{i\omega t}) \\ &= 2\sqrt{\frac{\hbar}{2m\omega}} |c_0 c_1| \cos(\omega t - \delta), \quad \delta = \arg(c_0^* c_1). \end{aligned}$$

The mean oscillation is a coherence effect between neighboring levels. If one coefficient vanishes, the amplitude vanishes and δ is irrelevant. A statistical mixture with the same populations lacks this cross term.

Figure/correspondence lesson: a highly excited number state has a rapidly oscillating spatial probability density, whose coarse-grained envelope approaches the classical distribution at the same energy. It remains stationary: large n alone does not produce a localized packet traveling around an orbit.

3.3.5 Coherent states

Define $a|\alpha\rangle = \alpha|\alpha\rangle$, $\alpha \in \mathbb{C}$; a is not Hermitian, so its eigenvalue need not be real. Write $|\alpha\rangle = \sum_n c_n |n\rangle$. Equating coefficients gives $c_{n+1}\sqrt{n+1} = \alpha c_n$, hence $c_n = c_0 \alpha^n / \sqrt{n!}$; normalization gives $|c_0| = e^{-|\alpha|^2/2}$.

$$\begin{aligned} \boxed{|\alpha\rangle &= e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle}, \\ P_n &= e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!}, \quad \bar{n} = \langle N \rangle = |\alpha|^2, \quad \langle H \rangle = \hbar\omega(|\alpha|^2 + \tfrac{1}{2}). \end{aligned}$$

The occupation distribution is Poisson. **Phase matters:** $|\alpha| = \sqrt{\bar{n}}$ always, whereas $\alpha = \sqrt{\bar{n}}$ requires *nonnegative* real α . Merely saying “real” (as in the source) misses the negative-real possibility.

Displacement construction:

$$|\alpha\rangle = D(\alpha)|0\rangle, \quad D(\alpha) = \exp(\alpha a^\dagger - \alpha^* a).$$

The exponent is anti-Hermitian, so D is unitary. Real and imaginary parts set the position and momentum displacements, respectively. Multiplying each number-state coefficient by its energy phase gives

$$\boxed{|\alpha, t\rangle = e^{-i\omega t/2} |\alpha e^{-i\omega t}\rangle}.$$

For $\alpha_t = \alpha e^{-i\omega t}$, ladder algebra yields

$$\begin{aligned} \langle x \rangle_t &= \sqrt{\frac{2\hbar}{m\omega}} \operatorname{Re} \alpha_t, & \langle p \rangle_t &= \sqrt{2m\hbar\omega} \operatorname{Im} \alpha_t, \\ (\Delta x)^2 &= \frac{\hbar}{2m\omega}, & (\Delta p)^2 &= \frac{m\hbar\omega}{2}, \quad \Delta x \Delta p = \frac{\hbar}{2}. \end{aligned}$$

The packet keeps its shape and its center follows a classical orbit, while nonzero vacuum fluctuations remain. The global factor $e^{-i\omega t/2}$ changes no probabilities. Coherent states are special superpositions, not energy eigenstates except for $\alpha = 0$.

3.4 Coordinate-space solution of the harmonic oscillator

Source PDF pp. 56–57.

Boundary-value problem: on the whole real line, require a square-integrable wave function solving

$$\left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2 \right] \psi = E \psi.$$

Nondimensionalization: with $x_0 = \sqrt{\hbar/(m\omega)}$, $\xi = x/x_0$ and $\lambda = 2E/(\hbar\omega)$,

$$(-\partial_\xi^2 + \xi^2)\psi = \lambda\psi.$$

Asymptotics: for $|\xi| \gg 1$, the leading exponential behaviors are $e^{\pm\xi^2/2}$; normalizability excludes the growing branch. The allowed solution can have an additional polynomial factor, so do not impose a pure Gaussian for every level. Put $\psi = H(\xi)e^{-\xi^2/2}$; then

$$\begin{aligned} \psi'' &= e^{-\xi^2/2} [H'' - 2\xi H' + (\xi^2 - 1)H], \\ H'' - 2\xi H' + (\lambda - 1)H &= 0. \end{aligned}$$

Power series and termination: insert $H = \sum_{r=0}^{\infty} a_r \xi^r$ and align powers:

$$(r+2)(r+1)a_{r+2} + (\lambda - 2r - 1)a_r = 0, \quad \boxed{a_{r+2} = \frac{2r+1-\lambda}{(r+2)(r+1)} a_r.}$$

Even and odd coefficients are separate chains, seeded by a_0 and a_1 . A nonterminating chain generically supplies an e^{ξ^2} factor in H , hence a nonnormalizable $e^{+\xi^2/2}$ branch in ψ ; decay at only one end is insufficient. The acceptable full-line solution is a terminating parity chain; the other chain is set to zero. Termination at degree n requires

$$\boxed{\lambda = 2n + 1 \implies E_n = \hbar\omega(n + \tfrac{1}{2}), \quad n = 0, 1, \dots}$$

The surviving polynomial solves the Hermite equation,

$$H_n'' - 2\xi H_n' + 2nH_n = 0, \quad H_0 = 1, \quad H_1 = 2\xi, \quad H_2 = 4\xi^2 - 2.$$

With the standard Hermite normalization one recovers the normalized $\psi_n(x)$ in 3.3.3. **Exam logic:** the differential equation admits formal solutions at other energies; acceptable boundary conditions select the discrete spectrum. Nondimensionalization alone does not quantize energy.

Self-test for 3.1–3.4: reconstruct before looking at the answers

1. Why is the infinitesimal generator Hermitian? When may time ordering be removed?
2. Must a superposition evolve observably? What happens within one degenerate eigenspace?
3. For $H = \omega S_z$ and initial $|+x\rangle$, find $P(-x)$ and the sign of $\langle S_y \rangle$.
4. State the full Heisenberg equation and the two sufficient conditions for conservation.
5. Derive $[x(t), x(0)]$ for a free particle. Why does Ehrenfest not generally give one classical trajectory?
6. What forces integer oscillator occupation, and what are $\langle 2|x|0\rangle$ and $\Delta x \Delta p$ in $|2\rangle$?
7. Does a highly excited number state oscillate? What ingredient makes $c_0|0\rangle + c_1|1\rangle$ oscillate?
8. If $\alpha = i\beta$ with real $\beta > 0$, give $\bar{n}$, $\langle x(0) \rangle$, $\langle p(0) \rangle$, and the evolved coherent label.
9. At $\lambda = 5$, which series seed must vanish, what is a_2/a_0 , and what energy follows?

Compact answers. **1.** The $O(\delta t)$ term of $U^\dagger U - 1$ is $i(\Omega^\dagger - \Omega)\delta t$; pairwise commuting $H(t)$ permits dropping $\mathcal{T}$. **2.** No: observable oscillations require nonzero cross terms with unequal energies; one degenerate eigenspace acquires one global phase. **3.** $\sin^2(\omega t/2)$; $\langle S_y \rangle = (\hbar/2) \sin \omega t$ with the signed ω . **4.** $\dot{X}_H = (i/\hbar)[H_H, X_H] + (\partial_t X_S)_H$; zero commutator and zero explicit derivative. **5.** $[x(0) + tp(0)/m, x(0)] = -i\hbar t/m$; generally $\langle V'(x) \rangle \neq V'(\langle x \rangle)$. **6.** Positivity of N forces lowering to terminate at zero; 0 and $5\hbar/2$. **7.** Its mean does not; neighboring-level coherence $c_0^*c_1 \neq 0$ produces the oscillation. **8.** β^2 , 0, $\sqrt{2m\hbar\omega}\beta$, and $i\beta e^{-i\omega t}$ (plus the state's global phase). **9.** $a_1 = 0$, $a_2/a_0 = -2$, $E = 5\hbar\omega/2$; $a_4 = 0$ and $H \propto H_2$.

3.4.1 A related example: the linear potential $V(x) = k|x|$

Source continuation PDF p. 58.

For $k > 0$ on the full line,

$$-\frac{\hbar^2}{2m}\psi'' + k|x|\psi = E\psi, \quad x_0 = \left(\frac{\hbar^2}{2mk}\right)^{1/3}, \quad y = \frac{x}{x_0}, \quad \epsilon = \frac{E}{kx_0}.$$

The scale follows by balancing $\hbar^2/(2mx_0^2) = kx_0$, so $E \sim kx_0 \propto \hbar^{2/3}k^{2/3}m^{-1/3}$. The consistent dimensionless equation is

$$-\frac{d^2\psi}{dy^2} + |y|\psi = \epsilon\psi.$$

Source correction: the continuation PDF's Eq. (3.153) instead prints $d^2\psi/dy^2 - 2|y|\psi = -2\epsilon\psi$. Its factors of two conflict with the stated x_0 , ϵ , and ground energy. For $y > 0$, normalizability leaves $\psi \propto \text{Ai}(y - \epsilon)$ and excludes Bi. Parity imposes

$$\text{Ai}'(-\epsilon) = 0 \quad (\text{even}), \quad \text{Ai}(-\epsilon) = 0 \quad (\text{odd}).$$

The first zero of Ai' gives $E_0 = 1.018792 \dots kx_0$. Both ψ and ψ' are continuous at $x = 0$: V has a cusp but no delta function. Dimensional analysis fixes the scale, not the Airy-zero coefficient.

3.5 The wave function, probability current, and classical limit

Source continuation PDF pp. 58–60.

3.5.1 Probability density and continuity equation

For $H = -\hbar^2\nabla^2/(2m) + V(\mathbf{r}, t)$ with real V , set $\rho = |\psi|^2$. Multiply the Schrödinger equation by ψ^* , its conjugate by ψ , and subtract:

$$\begin{aligned} \partial_t \rho &= \frac{i\hbar}{2m}(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) = -\nabla \cdot \mathbf{j}, \\ \mathbf{j} &= \left[\frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi) = \frac{1}{m} \text{Re}[\psi^* (-i\hbar \nabla) \psi] \right]. \end{aligned}$$

Thus $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$ and $d \int_{\Omega} \rho d^3r / dt = -\oint_{\partial\Omega} \mathbf{j} \cdot d\mathbf{S}$. Total probability is conserved when the boundary flux vanishes. For an effective complex potential $V = V_R + iV_I$,

$$\partial_t \rho + \nabla \cdot \mathbf{j} = \frac{2V_I}{\hbar} \rho;$$

$V_I < 0$ is a sink. This phenomenologically removes unresolved channels and is not unitary closed-system dynamics. The displayed $\mathbf{j}$ is probability current; electric current is $q\mathbf{j}$.

3.5.2 Phase and probability flow

Where $\rho > 0$, write

$$\psi = \sqrt{\rho} e^{iS/\hbar}, \quad \mathbf{j} = \rho \frac{\nabla S}{m} \equiv \rho \mathbf{v}.$$

Phase gradients drive flow. The field $\mathbf{v}$ is not automatically the trajectory velocity of an individual particle. At nodes this polar form is singular and S may be multivalued.

3.5.3 Quantum Hamilton–Jacobi equation and classical limit

Substitution of $\sqrt{\rho} e^{iS/\hbar}$ and separation into imaginary and real parts gives the exact pair

$$\begin{aligned} \partial_t \rho + \nabla \cdot \left(\rho \frac{\nabla S}{m} \right) &= 0, \\ \partial_t S + \frac{|\nabla S|^2}{2m} + V + Q &= 0, \quad Q = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \end{aligned}$$

The first transports amplitude; the second transports phase. Classical Hamilton–Jacobi theory follows where Q (and its relevant gradients) is negligible relative to the other scales. This is a slow-amplitude/WKB condition, not merely the slogan $\hbar \rightarrow 0$; it can fail near nodes, turning points, sharp boundaries, and interference fringes. For a stationary state, $S(\mathbf{r}, t) = W(\mathbf{r}) - Et$.

3.6 Propagators and path integrals

Source continuation PDF pp. 60–65.

Unless stated otherwise, take one spatial dimension, $t > t_0$, and $H = p^2/(2m) + V(x)$ with real, sufficiently regular V .

3.6.1 Propagator: one kernel evolves every state

$$K(x, t; x', t_0) = \langle x | U(t, t_0) | x' \rangle, \quad \boxed{\psi(x, t) = \int dx' K(x, t; x', t_0) \psi(x', t_0)}.$$

For time-independent H ,

$$K = \sum_n \psi_n(x) \psi_n^*(x') e^{-iE_n(t-t_0)/\hbar},$$

with continuum integrals added when required. Three operator identities become three reconstruction checks:

$$\begin{aligned} \lim_{t \rightarrow t_0^+} K(x, t; x', t_0) &= \delta(x - x') \quad \text{distributionally,} \\ K(x_2, t_2; x_0, t_0) &= \int dx_1 K(x_2, t_2; x_1, t_1) K(x_1, t_1; x_0, t_0), \\ i\hbar \partial_t K(x, t; x', t_0) &= H_x K(x, t; x', t_0). \end{aligned}$$

For $K_R = \Theta(t - t_0)K$,

$$(i\hbar \partial_t - H_x)K_R = i\hbar \delta(t - t_0) \delta(x - x').$$

Some texts instead define $G_R = -(i/\hbar)K_R$, whose right-hand side is simply $\delta(t - t_0)\delta(x - x')$; do not mix the two normalizations.

3.6.2 Free and harmonic-oscillator propagators

Insert momentum completeness, with $\langle x | p \rangle = e^{ipx/\hbar}/\sqrt{2\pi\hbar}$ and $T = t - t_0 > 0$:

$$\begin{aligned} K_0(x, t; x', t_0) &= \int \frac{dp}{2\pi\hbar} e^{\frac{i}{\hbar}[p(x-x') - p^2 T/(2m)]} \\ &= \boxed{\left(\frac{m}{2\pi i\hbar T}\right)^{1/2} e^{\frac{im(x-x')^2}{2\hbar T}}} = \left(\frac{m}{2\pi i\hbar T}\right)^{1/2} e^{iS_{\text{cl}}/\hbar}, \end{aligned}$$

where $S_{\text{cl}} = m(x - x')^2/(2T)$. The prefactor fixes dimensions, normalization, composition, and the delta-function limit; $e^{iS_{\text{cl}}/\hbar}$ alone is not the propagator. For comparison,

$$K_{\text{ho}}(x'', t; x', t_0) = \left(\frac{m\omega}{2\pi i\hbar \sin \omega T}\right)^{1/2} \exp\left\{\frac{im\omega}{2\hbar \sin \omega T}[(x''^2 + x'^2) \cos \omega T - 2x''x']\right\}.$$

Continue the square-root phase consistently through the caustics $\sin \omega T = 0$; the kernel there is distributional. The $\omega \rightarrow 0$ limit recovers K_0 .

3.6.3 Trace and spectral poles

For a discrete spectrum, with convergence/regulation understood,

$$G(t) = \int dx K(x, t; x, 0) = \text{Tr } e^{-iHt/\hbar} = \sum_n e^{-iE_n t/\hbar},$$

and

$$\boxed{\tilde{G}(E) = -\frac{i}{\hbar} \int_0^\infty dt e^{i(E+i0^+)t/\hbar} G(t) = \sum_n \frac{1}{E - E_n + i0^+}}.$$

Thus energy eigenvalues are poles. In infinite volume or with a continuum the trace may diverge and needs a box, density-of-states normalization, imaginary time, or another regulator.

3.6.4 Why K is an amplitude

In Heisenberg notation $K(x, t; x', t_0) = \langle x, t | x', t_0 \rangle$. It is an amplitude density, not a probability. Indistinguishable alternatives are added before taking $|\cdot|^2$; summing path probabilities would erase interference.

3.6.5 Time slicing: from composition to the path integral

Divide $T = t_f - t_i = N\varepsilon$, set $x_0 = x_i$, $x_N = x_f$, and insert $N - 1$ position identities:

$$K = \int \prod_{j=1}^{N-1} dx_j \prod_{j=0}^{N-1} \langle x_{j+1} | e^{-iH\varepsilon/\hbar} | x_j \rangle.$$

For $H = p^2/(2m) + V(x)$, a symmetric splitting (or equivalent consistent midpoint prescription) and a momentum Gaussian give

$$\begin{aligned} \langle x_{j+1} | e^{-iH\varepsilon/\hbar} | x_j \rangle &\simeq \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \exp \left\{ \frac{i\varepsilon}{\hbar} \left[\frac{m}{2} \left(\frac{x_{j+1} - x_j}{\varepsilon} \right)^2 - V(\bar{x}_j) \right] \right\}, \\ K &= \lim_{N \rightarrow \infty} \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{N/2} \int \prod_{j=1}^{N-1} dx_j e^{iS_N/\hbar} \equiv \int_{x_i}^{x_f} \mathcal{D}x e^{iS[x]/\hbar}, \\ S[x] &= \int_{t_i}^{t_f} dt \left[\frac{1}{2} m \dot{x}^2 - V(x) \right]. \end{aligned}$$

This limit *defines* the formal $\mathcal{D}x$, including its normalization. The source first displays the endpoint-ordered first-order Trotter product and then invokes the usual symmetric midpoint prescription; these are equivalent in the continuum for a regular coordinate-only V , but midpoint discretization should be tied to symmetric splitting when ordering matters.

The integral sums amplitudes over all fixed-endpoint histories; it does not say the particle secretly takes one classical trajectory. When stationary phase is controlled, rapidly varying phases away from $\delta S = 0$ cancel while nearby paths add coherently. Include every stationary path and its fluctuation phase; for quadratic actions the Gaussian semiclassical evaluation is exact. **Pitfalls:** never omit the short-time normalization, promote a Trotter approximation to an exact finite- ε identity, or assume typical real-time paths are differentiable classical curves. Velocity-dependent/gauge couplings require a specified ordering and discretization.

3.6.6 Recovering the Schrödinger equation

For one short step put $\xi = x - x'$ and expand:

$$\begin{aligned} \psi(x, t + \varepsilon) &= \left(\frac{m}{2\pi i \hbar \varepsilon} \right)^{1/2} \int d\xi e^{im\xi^2/(2\hbar\varepsilon)} \left(1 - \frac{i\varepsilon}{\hbar} V(x) + \dots \right) \\ &\times \left(\psi - \xi \partial_x \psi + \frac{\xi^2}{2} \partial_x^2 \psi + \dots \right). \end{aligned}$$

The normalized oscillatory Gaussian has $\langle \xi \rangle = 0$ and $\langle \xi^2 \rangle = i\hbar\varepsilon/m$. Therefore $\xi \sim \sqrt{\varepsilon}$: the quadratic term is $O(\varepsilon)$ and must be retained. Comparing terms gives

$$\boxed{i\hbar \partial_t \psi = \left(-\frac{\hbar^2}{2m} \partial_x^2 + V \right) \psi.}$$

The path integral is thus a reformulation of the same dynamics, not a new theory.

3.7 Gauge transformations and the phase of the wave function

Source continuation PDF pp. 65–69; SI conventions.

3.7.1 Constant potential versus observable relative phase

If $H' = H + V_0$ everywhere, then $|\psi'(t)\rangle = e^{-iV_0(t-t_0)/\hbar} |\psi(t)\rangle$, a global phase. But two coherent arms exposed to V_I, V_{II} for time T acquire

$$\boxed{\Delta\phi = -\frac{(V_I - V_{II})T}{\hbar}},$$

and the interference contains $1 + \cos \Delta\phi$. The energy zero is arbitrary only when shifted identically for all coherent alternatives; force-free does not mean phase-free.

3.7.2 Electromagnetic gauge covariance

$$\begin{aligned}\mathbf{B} &= \nabla \times \mathbf{A}, & \mathbf{E} &= -\nabla\phi - \partial_t \mathbf{A}, \\ \mathbf{A}' &= \mathbf{A} + \nabla\Lambda, & \phi' &= \phi - \partial_t \Lambda, \\ L &= \frac{1}{2}m\dot{\mathbf{r}}^2 + q\dot{\mathbf{r}} \cdot \mathbf{A} - q\phi, & \mathbf{p}_{\text{can}} &= m\dot{\mathbf{r}} + q\mathbf{A}, & \boldsymbol{\Pi} &= \mathbf{p} - q\mathbf{A}, \\ H &= \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} + q\phi, & [\Pi_i, \Pi_j] &= i\hbar q \epsilon_{ijk} B_k.\end{aligned}$$

Retain the operator order in $(\mathbf{p} - q\mathbf{A})^2$ because $\mathbf{p}$ does not generally commute with $\mathbf{A}(\mathbf{r}, t)$. Gauge covariance requires

$$\boxed{\psi' = e^{iq\Lambda/\hbar}\psi}, \quad (\mathbf{p} - q\mathbf{A}')\psi' = e^{iq\Lambda/\hbar}(\mathbf{p} - q\mathbf{A})\psi.$$

Hence

$$\mathbf{j} = \frac{1}{m} \text{Re}(\psi^* \boldsymbol{\Pi} \psi) = \frac{\hbar}{m} \text{Im}(\psi^* \nabla \psi) - \frac{q}{m} \mathbf{A} |\psi|^2 = \frac{\rho}{m} (\nabla S - q\mathbf{A}).$$

The diamagnetic term is essential; neither ∇S nor $\mathbf{A}$ alone is gauge invariant. For $\mathbf{B} = B\hat{\mathbf{z}}$, the symmetric gauge $\mathbf{A} = \frac{B}{2}(-y, x, 0)$ and Landau gauge $\mathbf{A} = (-By, 0, 0)$ differ by $\nabla(Bxy/2)$ and predict identical observables.

3.7.3 Aharonov–Bohm phase

Two coherent paths can lie where $\mathbf{B} = 0$ yet enclose flux. The action term $q \int_C \mathbf{A} \cdot d\boldsymbol{\ell}$ gives the oriented relative phase

$$\boxed{\Delta\phi = \frac{q}{\hbar} \oint \mathbf{A} \cdot d\boldsymbol{\ell} = \frac{q}{\hbar} \int \mathbf{B} \cdot d\mathbf{S} = \frac{q\Phi_B}{\hbar}}.$$

Only the loop phase/flux, not $\mathbf{A}$ at one point, is gauge invariant. Interference periodicity gives

$$\boxed{\Phi_0 = \frac{h}{|q|}}; \quad \frac{h}{e} \text{ for one electron, } \frac{h}{2e} \text{ for a charge-}2e \text{ condensate.}$$

The effect requires coherence and enclosed flux; it is not a local Lorentz-force effect along either field-free arm.

3.7.4 Magnetic monopole and Dirac quantization

Assume a monopole with the source convention $\Phi_M = \oint \mathbf{B} \cdot d\mathbf{S} = 4\pi g$ and $\mathbf{B} = g\hat{\mathbf{r}}/r^2$. Nonzero flux forbids one global regular vector potential. Use north/south patches:

$$\mathbf{A}_N = g \frac{1 - \cos\theta}{r \sin\theta} \hat{\boldsymbol{\varphi}}, \quad \mathbf{A}_S = -g \frac{1 + \cos\theta}{r \sin\theta} \hat{\boldsymbol{\varphi}}, \quad \mathbf{A}_N - \mathbf{A}_S = \nabla(2g\varphi).$$

On the overlap, $\psi_N = e^{i2gq\varphi/\hbar} \psi_S$. Single-valuedness under $\varphi \mapsto \varphi + 2\pi$ requires

$$e^{i4\pi qg/\hbar} = 1 \implies \boxed{qg = \frac{n\hbar}{2}, \quad n \in \mathbb{Z}} \iff \boxed{q\Phi_M = n\hbar}.$$

The numerical form depends on the definition of g ; retain $\Phi_M = 4\pi g$. This is a conditional consistency result, not evidence that monopoles exist.

3.8 Examples of one-particle dynamics

Source continuation PDF pp. 69–73.

3.8.1 One-dimensional square wells

Infinite centered well: $V = 0$ for $|x| < a$ and $V = \infty$ outside. With $u(\pm a) = 0$ and $k = \sqrt{2mE}/\hbar$,

$$\begin{aligned}u_+(x) &\propto \cos kx, & ka &= (r + \tfrac{1}{2})\pi, & r &= 0, 1, \dots, \\ u_-(x) &\propto \sin kx, & ka &= r\pi, & r &= 1, 2, \dots,\end{aligned}$$

which interleave into

$$E_n = \frac{n^2 \pi^2 \hbar^2}{8ma^2}, \quad n = 1, 2, \dots$$

The width is $2a$; treating a as the width causes a factor-of-four error.

Finite centered well: $V = 0$ for $|x| < a$ and $V = V_0 > 0$ outside. For $0 < E < V_0$ define

$$k = \frac{\sqrt{2mE}}{\hbar}, \quad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}, \quad k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}.$$

Continuity of u, u' at $x = a$ yields

$$k \tan ka = \kappa \quad (\text{even}), \quad -k \cot ka = \kappa \quad (\text{odd}).$$

There are finitely many bound states. Their $e^{-\kappa|x|}$ tails lengthen as $E \rightarrow V_0^-$. For $E > V_0$ the exterior is oscillatory and belongs to the scattering continuum; $V_0 \rightarrow \infty$ recovers the infinite well. If $V(x) = V(-x)$ then $[H, P] = 0$, $P^2 = 1$, and bound states may be chosen even or odd. Under the usual one-dimensional regularity assumptions, bound levels are nondegenerate; alternation additionally follows from node ordering.

3.8.2 Uniform magnetic field: Landau levels

Let a spinless charge q move in the xy plane with $\mathbf{B} = B\hat{\mathbf{z}}$, $B > 0$, and no scalar potential: $H = \mathbf{\Pi}^2/(2m)$, $\omega_c = |q|B/m$. In Landau gauge $\mathbf{A} = (-By, 0, 0)$, $[p_x, H] = 0$. For $\psi = e^{ikx}\phi(y)$,

$$H\phi = \left[\frac{p_y^2}{2m} + \frac{1}{2}m\omega_c^2(y - y_0)^2 \right] \phi, \quad y_0 = -\frac{\hbar k}{qB},$$

so

$$E_n = \hbar\omega_c(n + \tfrac{1}{2}), \quad N_{\text{deg}} = \frac{A|q|B}{h} = \frac{\Phi_B}{h/|q|}$$

(up to edge corrections). The conserved k , equivalently the guiding center y_0 , labels the macroscopic degeneracy. In symmetric gauge let $s = \text{sgn}(qB)$ and

$$a = \frac{\Pi_x + is\Pi_y}{\sqrt{2\hbar|q|B}}, \quad a^\dagger = \frac{\Pi_x - is\Pi_y}{\sqrt{2\hbar|q|B}}.$$

Then $[a, a^\dagger] = 1$ and $H = \hbar\omega_c(a^\dagger a + \frac{1}{2})$. The factor s is essential for a negative charge. Different gauges change labels and phases, not the spectrum or degeneracy. The displayed result neglects spin/Zeman energy, interactions, disorder, confinement, and edge dispersion; in three dimensions add $p_z^2/(2m)$.

3.9 Chapter synthesis

Source continuation PDF pp. 73–74.

- Amplitude and phase obey continuity plus an exact quantum Hamilton–Jacobi equation; Q identifies where classical reasoning fails.
- $K = \langle x|U|x' \rangle$ evolves every state. Composition and time slicing yield $\int \mathcal{D}x e^{iS/\hbar}$; stationary-phase cancellation recovers classical action without discarding quantum paths.
- Gauge transformations change potentials and wave-function phase together, leaving mechanical momentum, current, loop phase, and flux invariant. Aharonov–Bohm interference and Dirac quantization reveal global phase structure.
- Wells and Landau levels exemplify the reusable chain: exploit symmetry or gauge choice, reduce the equation, impose boundary/normalizability conditions, and then read off spectrum and degeneracy.

Self-test for 3.4.1–3.9

1. For $V = k|x|$, derive x_0 and the even/odd Airy conditions. Why is the printed Eq. (3.153) inconsistent?
2. For $V = V_R + iV_I$, which sign of V_I absorbs probability?
3. From $\psi = \sqrt{\rho}e^{iS/\hbar}$ identify $\mathbf{j}$ and Q . Name two places where neglecting Q can fail.
4. State the propagator's initial, composition, and evolution equations. What part of K_0 is not fixed by $e^{iS_{cl}/\hbar}$?
5. Why does the $\xi^2 \partial_x^2 \psi$ term survive the short-time expansion?

6. How do $(\mathbf{A}, \phi, \psi)$ transform, and which phase-gradient combination is gauge invariant?
7. Give the Aharonov–Bohm phase and flux period. Derive Dirac quantization from the north–south transition function.
8. Write the two finite-well bound-state conditions and the Landau energy and degeneracy per area.

Compact answers. **1.** Balance $\hbar^2/(2mx_0^2) = kx_0$; $\text{Ai}'(-\epsilon) = 0$ (even), $\text{Ai}(-\epsilon) = 0$ (odd); the printed factors of two conflict with this x_0 and ϵ . **2.** $V_I < 0$, since the source is $2V_I\rho/\hbar$. **3.** $\mathbf{j} = \rho\nabla S/m$ and $Q = -(\hbar^2/2m)\nabla^2\sqrt{\rho}/\sqrt{\rho}$; e.g. nodes and turning points. **4.** $K \rightarrow \delta$, convolution, and $i\hbar\partial_t K = H_x K$; the normalization and branch/Maslov phase are additional. **5.** Oscillatory-Gaussian scaling gives $\langle \xi^2 \rangle = i\hbar\epsilon/m = O(\epsilon)$. **6.** $\mathbf{A}' = \mathbf{A} + \nabla\Lambda$, $\phi' = \phi - \partial_t\Lambda$, $\psi' = e^{iq\Lambda/\hbar}\psi$; $\nabla S - q\mathbf{A}$ is invariant. **7.** $\Delta\phi = q\Phi_B/\hbar$, $\Phi_0 = h/|q|$; single-valued $e^{i2qg\varphi/\hbar}$ gives $qg = n\hbar/2$ or $q\Phi_M = nh$. **8.** $k \tan ka = \kappa$, $-k \cot ka = \kappa$; $E_n = \hbar|q|B(n + 1/2)/m$ and $N_{\text{deg}}/A = |q|B/h$.

Chapter 4: Angular Momentum and Rotations

Scope. Sections 4.1–4.8, lecture pp. 75–100 (PDF pp. 85–110; file page = lecture page +10). The first part below covers 4.1–4.3 on lecture pp. 75–85. Spherical harmonics are an explicitly marked reconstruction supplement, not a separate heading in the source.

4.1 Rotations and angular momentum

Source: lecture pp. 75–80; PDF pp. 85–90.

4.1.1 Finite and infinitesimal rotations

Convention: active, right-handed rotations of column vectors, $\mathbf{r}' = R\mathbf{r}$; products act rightmost first. Proper rotations satisfy $R^T R = I$, $\det R = 1$.

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad R_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix},$$

$$R_z(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Same-axis rotations add angles and commute; different-axis rotations generally do not. Define $G_i = \partial_\theta R_i(0)$, so $R_i(\epsilon) = I + \epsilon G_i + O(\epsilon^2)$ and $G_i \mathbf{r} = \hat{\mathbf{e}}_i \times \mathbf{r}$. Explicitly,

$$G_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad G_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad G_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Derivation backbone: $[G_i, G_j] = \epsilon_{ijk} G_k$ gives

$$R_x(\epsilon) R_y(\epsilon) R_x(-\epsilon) R_y(-\epsilon) = I + \epsilon^2 [G_x, G_y] + O(\epsilon^3) \\ = R_z(\epsilon^2) + O(\epsilon^3).$$

On states $|\psi\rangle_R = D|\psi\rangle$. Continuity and unitarity give a Hermitian generator:

$$D(\mathbf{n}, d\theta) = I - \frac{i}{\hbar} d\theta \mathbf{n} \cdot \mathbf{J} + O(d\theta^2), \quad \boxed{D(\mathbf{n}, \theta) = e^{-i\theta \mathbf{n} \cdot \mathbf{J}/\hbar}}.$$

The quantum group commutator is $I - \epsilon^2 [J_x, J_y]/\hbar^2 + O(\epsilon^3)$; equate it with $I - i\epsilon^2 J_z/\hbar$ to obtain

$$\boxed{[J_i, J_j] = i\hbar \epsilon_{ijk} J_k}.$$

Global qualification: composition is exact for a representation of $SU(2)$ (chosen lifts of spatial rotations). Half-integer spin gives a projective representation of $SO(3)$, not a globally single-valued ordinary $D(R)$; this does not alter the infinitesimal algebra.

4.1.2 Orbital angular momentum as the generator of spatial rotations

For a spinless scalar wave function, $D(R)|\mathbf{r}\rangle = |R\mathbf{r}\rangle$ implies

$$\psi_R(\mathbf{r}) = \psi(R^{-1}\mathbf{r}) = \psi(\mathbf{r} - \boldsymbol{\theta} \times \mathbf{r}) + O(\theta^2) \\ = [I - \boldsymbol{\theta} \cdot (\mathbf{r} \times \nabla)] \psi = \left[I - \frac{i}{\hbar} \boldsymbol{\theta} \cdot (\mathbf{r} \times \mathbf{p}) \right] \psi, \quad \mathbf{p} = -i\hbar \nabla.$$

Thus $\boxed{\mathbf{L} = \mathbf{r} \times \mathbf{p}}$, not a separate classical postulate.

$$L_x = -i\hbar(y\partial_z - z\partial_y), \quad L_y = -i\hbar(z\partial_x - x\partial_z), \quad L_z = -i\hbar(x\partial_y - y\partial_x).$$

With $[r_i, p_j] = i\hbar \delta_{ij}$, $[r_i, r_j] = [p_i, p_j] = 0$,

$$[r_a p_b, r_c p_d] = i\hbar(\delta_{ad} r_c p_b - \delta_{bc} r_a p_d) \implies [L_i, L_j] = i\hbar \epsilon_{ijk} L_k.$$

Symmetry and conservation: $D^\dagger H D = H \implies [H, J_i] = 0$ for all axes. With no explicit time dependence, $\dot{J}_i = (i/\hbar)[H, J_i] = 0$. For $H = \mathbf{p}^2/(2M) + V(r)$ the generator is $\mathbf{L}$. Axial symmetry alone conserves only the corresponding component; invariance under one isolated finite rotation is not full rotational symmetry.

4.1.3 Rotation of a spin-1/2 state

Spin is intrinsic; $\mathbf{S} = \hbar\boldsymbol{\sigma}/2$, $[S_i, S_j] = i\hbar\epsilon_{ijk}S_k$. Use $\sigma_i\sigma_j = \delta_{ij}I + i\epsilon_{ijk}\sigma_k$, so $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$ for $|\mathbf{n}| = 1$. Separating even and odd powers gives

$$U(\mathbf{n}, \theta) = I \cos(\theta/2) - i\mathbf{n} \cdot \boldsymbol{\sigma} \sin(\theta/2).$$

For $U_z = \text{diag}(e^{-i\theta/2}, e^{i\theta/2})$,

$$\begin{aligned} U_z^\dagger S_x U_z &= S_x \cos \theta - S_y \sin \theta, & U_z^\dagger S_y U_z &= S_x \sin \theta + S_y \cos \theta, \\ U_z^\dagger S_z U_z &= S_z, & U(\mathbf{n}, 2\pi) &= -I, \quad U(\mathbf{n}, 4\pi) = I. \end{aligned}$$

Hence $\langle \mathbf{S} \rangle_R = R\langle \mathbf{S} \rangle$: observables transform as vectors although spinors use half-angles. An isolated global minus sign changes no probability; a relative minus sign between coherently rotated and unrotated paths is observable.

4.1.4 Spin precession revisited

Fix the sign through the Hamiltonian, rather than absorbing it ambiguously:

$$\boldsymbol{\mu} = \gamma \mathbf{S}, \quad \mathbf{B} = B\hat{\mathbf{z}}, \quad H = -\gamma B S_z = \omega_L S_z, \quad \omega_L = -\gamma B.$$

For an electron $\gamma = -g\mu_B/\hbar$, hence $\omega_L = g\mu_B B/\hbar$ for signed B and positive g, μ_B .

$$\begin{aligned} U(t) &= e^{-i\omega_L t \sigma_z/2} = U_z(\omega_L t), \\ \langle S_x \rangle_t &= \langle S_x \rangle_0 \cos \omega_L t - \langle S_y \rangle_0 \sin \omega_L t, \\ \langle S_y \rangle_t &= \langle S_y \rangle_0 \cos \omega_L t + \langle S_x \rangle_0 \sin \omega_L t, & \langle S_z \rangle_t &= \langle S_z \rangle_0. \end{aligned}$$

The sign check is $\dot{S}_x = -\omega_L S_y$, $\dot{S}_y = \omega_L S_x$. For nonzero ω_L , the *operator* satisfies $U(2\pi/|\omega_L|) = -I$ and $U(4\pi/|\omega_L|) = I$. Generic transverse means have the first period; exact kets have the second with this energy-zero convention. Rays already return at the first, and a S_z eigenstate has a stationary ray and mean at all times.

4.2 Rotation groups and Euler angles

Source: lecture pp. 80–82; PDF pp. 90–92.

4.2.1 Basic group concepts

A group has closure, identity, inverses, and associativity; commutativity is *not* required. Abelian means $ab = ba$ for every pair. A representation preserves products:

$$D(g_1 g_2) = D(g_1) D(g_2), \quad D(e) = I, \quad D(g^{-1}) = D(g)^{-1}.$$

A unitary representation also has $D(g)^{-1} = D(g)^\dagger$. For quantum rays a projective representation is allowed: $D(R_1)D(R_2) = e^{i\chi(R_1, R_2)}D(R_1 R_2)$; choose $SU(2)$ lifts to remove this ambiguity for angular momentum. Different physical systems realize different representations, possibly with multiple copies of a given j .

4.2.2 $O(3)$, $SO(3)$, and the meaning of “special”

$$O(3) = \{R \in \mathbb{R}^{3 \times 3} : R^T R = I\}, \quad (\det R)^2 = 1, \quad SO(3) = \{R \in O(3) : \det R = 1\}.$$

“Special” means determinant one. $SO(3)$ is the identity-connected component; $-I$ is spatial inversion, with determinant -1 , and lies in $O(3) \setminus SO(3)$. The three real antisymmetric G_i determine the local Lie algebra; $-iJ_i/\hbar$ obey the same commutators. Local algebra alone does not determine the group’s global topology.

4.2.3 $SU(2)$ and the double cover of $SO(3)$

$$\begin{aligned} SU(2) &= \{U \in \mathbb{C}^{2 \times 2} : U^\dagger U = I, \det U = 1\}, \\ U &= \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \quad |a|^2 + |b|^2 = 1, \quad U = a_0 I - i\mathbf{a} \cdot \boldsymbol{\sigma}, \quad a_0, \mathbf{a} \text{ real}, \quad a_0^2 + |\mathbf{a}|^2 = 1. \end{aligned}$$

Set $a_0 = \cos(\theta/2)$, $\mathbf{a} = \mathbf{n} \sin(\theta/2)$. Define the covering map by

$$U(\mathbf{v} \cdot \boldsymbol{\sigma})U^\dagger = (R\mathbf{v}) \cdot \boldsymbol{\sigma}, \quad U \text{ and } -U \mapsto R.$$

Its kernel is $\{I, -I\}$, so $SO(3) \simeq SU(2)/\{\pm I\}$. $SU(2)$ is simply connected; $SO(3)$ is not. In a spin- j irreducible representation,

$$D^{(j)}(2\pi) = (-1)^{2j}I.$$

Integer j descends to an ordinary $SO(3)$ representation. Half-integer j is an ordinary $SU(2)$ representation but only a projective $SO(3)$ representation. Both obey the same angular-momentum algebra.

4.2.4 Euler rotations

Active fixed-axis z - y - z convention; γ acts first:

$$\begin{aligned} R(\alpha, \beta, \gamma) &= R_z(\alpha)R_y(\beta)R_z(\gamma), \\ D^{(j)}(\alpha, \beta, \gamma) &= e^{-i\alpha J_z/\hbar} e^{-i\beta J_y/\hbar} e^{-i\gamma J_z/\hbar}, \\ D^{(1/2)} &= \begin{pmatrix} e^{-i(\alpha+\gamma)/2} \cos(\beta/2) & -e^{-i(\alpha-\gamma)/2} \sin(\beta/2) \\ e^{i(\alpha-\gamma)/2} \sin(\beta/2) & e^{i(\alpha+\gamma)/2} \cos(\beta/2) \end{pmatrix}. \end{aligned}$$

Obtain the matrix by multiplying the diagonal z factors around $\cos(\beta/2)I - i\sin(\beta/2)\sigma_y$. It is unitary with determinant one. Euler coordinates are not unique, particularly at $\beta = 0, \pi$; spatial-angle identification must respect the lift sign for spinors.

4.3 Eigenvalues and eigenstates of angular momentum

Source: lecture pp. 82–85; PDF pp. 92–95.

Algebraic spectrum (unitary representation, normalizable ladder states):

$$\begin{aligned} J^2 &= J_x^2 + J_y^2 + J_z^2, \quad J_{\pm} = J_x \pm iJ_y, \quad J_{\pm}^{\dagger} = J_{\mp}, \\ [J^2, J_i] &= 0, \quad [J_z, J_{\pm}] = \pm\hbar J_{\pm}, \quad [J_+, J_-] = 2\hbar J_z, \\ J_- J_+ &= J^2 - J_z^2 - \hbar J_z, \quad J_+ J_- = J^2 - J_z^2 + \hbar J_z. \end{aligned}$$

Start with $J^2|a, b\rangle = a|a, b\rangle$, $J_z|a, b\rangle = b|a, b\rangle$. The commutators shift only b by $\pm\hbar$, and

$$\|J_{\pm}|a, b\rangle\|^2 = a - b(b \pm \hbar) \geq 0.$$

For fixed finite a , the allowed b are bounded; the ladder terminates at both ends:

$$\begin{aligned} a &= b_{\max}(b_{\max} + \hbar) = b_{\min}(b_{\min} - \hbar), \\ (b_{\max} + b_{\min})(b_{\max} - b_{\min} + \hbar) &= 0. \end{aligned}$$

Since $b_{\max} \geq b_{\min}$, the second factor is positive, so $b_{\min} = -b_{\max}$. With $b_{\max} - b_{\min} = N\hbar$, $N \in \{0, 1, \dots\}$, put $j = N/2$ and $b = m\hbar$:

$$J^2|jm\rangle = \hbar^2 j(j+1)|jm\rangle, \quad J_z|jm\rangle = \hbar m|jm\rangle, \quad j = 0, \frac{1}{2}, 1, \dots, \quad m = -j, -j+1, \dots, j.$$

Choose ladder phases to make nonzero coefficients positive:

$$J_{\pm}|jm\rangle = \hbar\sqrt{(j \mp m)(j \pm m + 1)}|j, m \pm 1\rangle.$$

A fixed- j irreducible multiplet has $2j+1$ states. Extra labels distinguish repeated multiplets. $|\mathbf{J}| = \hbar\sqrt{j(j+1)}$, not $j\hbar$; even $m = j$ has transverse fluctuations. The algebra permits both integer and half-integer j ; scalar orbital boundary conditions select integer l .

4.3.1 Matrix representations

Always use descending m : $(|j, j\rangle, |j, j-1\rangle, \dots, |j, -j\rangle)$. Row m' is output, column m is input:

$$\begin{aligned} (J_z)_{m'm} &= \hbar m \delta_{m'm}, \quad (J_{\pm})_{m'm} = \hbar\sqrt{j(j+1) - m(m \pm 1)} \delta_{m', m \pm 1}, \\ J_x &= (J_+ + J_-)/2, \quad J_y = (J_+ - J_-)/(2i). \end{aligned}$$

Thus J_+ is upper off-diagonal and $J_- = J_+^{\dagger}$. $j = 1/2$:

$$J_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad J_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

$j = 1$:

$$J_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad J_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

$j = 3/2$:

$$J_x = \frac{\hbar}{2} \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}, \quad J_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i\sqrt{3} & 0 & 0 \\ i\sqrt{3} & 0 & -2i & 0 \\ 0 & 2i & 0 & -i\sqrt{3} \\ 0 & 0 & i\sqrt{3} & 0 \end{pmatrix},$$

$$J_z = \frac{\hbar}{2} \text{diag}(3, 1, -1, -3).$$

Checks: Hermiticity, $[J_x, J_y] = i\hbar J_z$, and $J_x^2 + J_y^2 + J_z^2 = \hbar^2 j(j+1)I_{2j+1}$ reconstruct every coefficient and sign.

4.3.2 Wigner rotation matrices

Insert J_z eigenvalues on both sides of the middle Euler factor:

$$\boxed{D_{m'm}^{(j)}(\alpha, \beta, \gamma) = e^{-im'\alpha} d_{m'm}^{(j)}(\beta) e^{-im\gamma}},$$

$$d_{m'm}^{(j)}(\beta) = \langle jm' | e^{-i\beta J_y/\hbar} | jm \rangle, \quad d^{(1/2)}(\beta) = \begin{pmatrix} \cos(\beta/2) & -\sin(\beta/2) \\ \sin(\beta/2) & \cos(\beta/2) \end{pmatrix},$$

$$d^{(1)}(\beta) = \begin{pmatrix} \cos^2(\beta/2) & -\sin\beta/\sqrt{2} & \sin^2(\beta/2) \\ \sin\beta/\sqrt{2} & \cos\beta & -\sin\beta/\sqrt{2} \\ \sin^2(\beta/2) & \sin\beta/\sqrt{2} & \cos^2(\beta/2) \end{pmatrix}.$$

These small- d matrices are real orthogonal in this basis. Reconstruct from $d(0) = I$ and $\partial_\beta d = -(i/\hbar)J_y d$; consequently $d(-\beta) = d(\beta)^T$. A passive rotation, reverse basis order, or different state phases changes displayed signs; do not mix conventions. For initial $|jm\rangle$, a rotated J_z measurement has probability $|D_{m'm}^{(j)}|^2 = |d_{m'm}^{(j)}|^2$.

Reconstruction supplement: orbital eigenfunctions and spherical harmonics

Added to connect 4.1.2 with 4.3; not explicitly developed on lecture pp. 75–85. Use polar angle ϑ , azimuth φ , and $d\Omega = \sin\vartheta d\vartheta d\varphi$:

$$L_z = -i\hbar\partial_\varphi, \quad L^2 = -\hbar^2 \left[\frac{1}{\sin\vartheta} \partial_\vartheta (\sin\vartheta \partial_\vartheta) + \frac{1}{\sin^2\vartheta} \partial_\varphi^2 \right],$$

$$L_\pm = \hbar e^{\pm i\varphi} (\pm \partial_\vartheta + i \cot\vartheta \partial_\varphi).$$

$L_z Y = \hbar m Y \Rightarrow Y = \Theta(\vartheta) e^{im\varphi}$; scalar single-valuedness requires $m \in \mathbb{Z}$. With $x = \cos\vartheta$ and $L^2 Y = \hbar^2 \lambda Y$,

$$(1-x^2)\Theta'' - 2x\Theta' + \left[\lambda - \frac{m^2}{1-x^2} \right] \Theta = 0.$$

Regularity at both poles and square-integrability give $\lambda = l(l+1)$, $l = 0, 1, \dots$, $|m| \leq l$. For $m \geq 0$ define the Condon–Shortley phase in P_l^m itself:

$$P_l(x) = \frac{1}{2^l l!} \frac{d^l}{dx^l} (x^2 - 1)^l, \quad P_l^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m P_l(x)}{dx^m},$$

$$\boxed{Y_{lm}(\vartheta, \varphi) = \sqrt{\frac{2l+1}{4\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos\vartheta) e^{im\varphi}}, \quad Y_{l,-m} = (-1)^m Y_{lm}^*.$$

No additional $(-1)^m$ belongs in the boxed formula with this definition.

$$\int Y_{l'm'}^* Y_{lm} d\Omega = \delta_{ll'} \delta_{mm'}, \quad L_\pm Y_{lm} = \hbar \sqrt{(l \mp m)(l \pm m + 1)} Y_{l, m \pm 1},$$

$$Y_{00} = \frac{1}{\sqrt{4\pi}}, \quad Y_{10} = \sqrt{\frac{3}{4\pi}} \cos\vartheta, \quad Y_{1, \pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin\vartheta e^{\pm i\varphi}.$$

For a central potential, $\psi = R_l(r) Y_{lm}$; radial/energy labels remain independent of the angular multiplet. The bridge to Wigner matrices in the stated convention is

$$Y_{lm}(\vartheta, \varphi) = \sqrt{\frac{2l+1}{4\pi}} D_{m0}^{(l)*}(\varphi, \vartheta, 0), \quad Y_{lm}(R^{-1}\hat{\mathbf{r}}) = \sum_{m'} Y_{lm'}(\hat{\mathbf{r}}) D_{m'm}^{(l)}(R).$$

Source caveats and corrections

- **Global representation omission** (lecture pp. 77, 80–81; PDF pp. 87, 90–91): the source’s unqualified product-preserving $D(R)$ description must be read locally or with $SU(2)$ lifts. It is not an ordinary global $SO(3)$ representation for half-integer spin; the notes make the projective qualification explicit.
- **Overstated return-time claim** (lecture p. 80; PDF p. 90): “the spin expectation value returns after $2\pi/\omega_L$, whereas the spinor itself returns only after $4\pi/\omega_L$ ” is not a universal statement about a minimal physical period. S_z eigenstates have stationary means and rays; exact-ket phases depend on the energy-zero convention. The $2\pi/4\pi$ operator identities are correct; periods use $|\omega_L|$.
- No definite algebraic or matrix-sign error was found in 4.1–4.3. The extracted $j = 3/2$ and spin-1 small- d matrices have broken line ordering; their reconstructed forms satisfy the generator equations and Casimir checks. The p. 85 running header mentions 4.4, but 4.3.2 is still on that page; 4.4 content is excluded.

Self-test for 4.1–4.3: reconstruct before looking at the answers

1. Why is the rotation generator Hermitian, and how does the small group commutator fix $[J_x, J_y]$?
2. Why does an active scalar rotation evaluate ψ at $R^{-1}\mathbf{r}$? Which operator follows?
3. Rotate $|+x\rangle$ through θ about z : find its components and transverse spin means. What changes at 2π ?
4. For $H = \omega_L S_z$, what is the signed precession direction? Does every state have an observable $2\pi/|\omega_L|$ period?
5. Distinguish $O(3)$, $SO(3)$, and $SU(2)$; when does spin j give an ordinary $SO(3)$ representation?
6. In the Euler product, which factor acts first? Derive the phases multiplying $d_{m'm}^{(j)}$.
7. Derive $b_{\min} = -b_{\max}$ and $2j \in \mathbb{Z}_{\geq 0}$ from endpoint norms, not by assuming the spectrum.
8. Compute $J_+|\frac{3}{2}, -\frac{1}{2}\rangle$, and $\langle 1, 0|J_y|1, 1\rangle$. Find $\langle J_x^2 + J_y^2 \rangle$ in $|j, j\rangle$.
9. A spin-1 state $|1, 1\rangle$ is rotated through β about y . Give the probabilities of $m' = 1, 0, -1$.
10. What excludes half-integer orbital l for a scalar wave function? Give $Y_{1,-1}$ and the convention relating Y_{lm} to $D^{(l)}$.

Compact answers. **1.** $D^\dagger D = I$ to first order requires $J^\dagger = J$; $-\epsilon^2[J_x, J_y]/\hbar^2 = -i\epsilon^2 J_z/\hbar$. **2.** The rotated amplitude at $\mathbf{r}$ came from $R^{-1}\mathbf{r}$; Taylor expansion gives $\mathbf{L} = \mathbf{r} \times (-i\hbar\nabla)$. **3.** $(e^{-i\theta/2}, e^{i\theta/2})^T/\sqrt{2}$; $(\langle S_x \rangle, \langle S_y \rangle) = \frac{\hbar}{2}(\cos\theta, \sin\theta)$. At 2π the ket changes sign but its ray and means do not. **4.** $\langle \mathbf{S} \rangle = \omega_L \hat{\mathbf{z}} \times \langle \mathbf{S} \rangle$; no, a S_z eigenstate has stationary means. **5.** Orthogonal matrices, determinant-one orthogonal matrices, and determinant-one unitary 2×2 matrices; only integer j descends, since $(-1)^{2j} = 1$. **6.** The rightmost z rotation (γ); $e^{-im'\alpha}$ from the bra and $e^{-im\gamma}$ from the ket. **7.** Endpoint equality factors as $(b_{\max} + b_{\min})(b_{\max} - b_{\min} + \hbar) = 0$; the second factor is positive. A finite number N of steps gives $2b_{\max} = N\hbar$ and $j = N/2$. **8.** $2\hbar|\frac{3}{2}, \frac{1}{2}\rangle$; $i\hbar/\sqrt{2}$; $\hbar^2[j(j+1) - j^2] = j\hbar^2$. **9.** $\cos^4(\beta/2)$, $\sin^2 \beta/2$, $\sin^4(\beta/2)$, respectively; they sum to one. **10.** Single-valuedness gives integer m and regular poles give integer l ; $Y_{1,-1} = \sqrt{3/(8\pi)} \sin\vartheta e^{-i\varphi}$; $Y_{lm} = \sqrt{(2l+1)/(4\pi)} D_{m0}^{(l)*}(\varphi, \vartheta, 0)$. **Scope and conventions.** Sections 4.4–4.5, lecture pp. 85–91 (PDF pp. 95–101); exclude the preceding 4.3.2 and following 4.6. Rotations are *active*, $U(\mathbf{n}, \theta) = e^{-i\theta \mathbf{n} \cdot \mathbf{J}/\hbar}$; rows and columns of spin matrices are ordered $m = j, j-1, \dots, -j$. Single-spin ladders have positive real matrix elements, $J_\pm|jm\rangle = \hbar\sqrt{(j \mp m)(j \pm m + 1)}|j, m \pm 1\rangle$.

4.4 The Schwinger oscillator representation

Source: lecture pp. 85–86; PDF pp. 95–96.

Mechanism: two ordinary boson modes encode spin as their occupation imbalance; transferring one boson changes m without changing j . Assume independent canonical modes,

$$[a_\sigma, a_{\sigma'}^\dagger] = \delta_{\sigma\sigma'}, \quad [a_\sigma, a_{\sigma'}] = [a_\sigma^\dagger, a_{\sigma'}^\dagger] = 0, \quad \sigma = +, -.$$

Define

$$\begin{aligned} N_\pm &= a_\pm^\dagger a_\pm, \quad N = N_+ + N_-, \quad J_+ = \hbar a_+^\dagger a_-, \quad J_- = \hbar a_-^\dagger a_+, \\ J_z &= \frac{\hbar}{2}(N_+ - N_-), \quad J_x = \frac{J_+ + J_-}{2}, \quad J_y = \frac{J_+ - J_-}{2i}. \end{aligned}$$

Algebra and Casimir derivation: use $[N_\sigma, a_\sigma^\dagger] = a_\sigma^\dagger$ and $[N_\sigma, a_\sigma] = -a_\sigma$:

$$\begin{aligned} [J_z, J_+] &= \frac{\hbar^2}{2}(a_+^\dagger a_- + a_+^\dagger a_-) = \hbar J_+, \\ J_+ J_- &= \hbar^2 N_+(N_- + 1), \quad J_- J_+ = \hbar^2 N_-(N_+ + 1), \\ [J_+, J_-] &= \hbar^2(N_+ - N_-) = 2\hbar J_z, \quad [J_z, J_-] = -\hbar J_-, \quad [N, J_i] = 0, \\ \frac{\mathbf{J}^2}{\hbar^2} &= \frac{(N_+ - N_-)^2}{4} + \frac{N_+(N_- + 1) + N_-(N_+ + 1)}{2} = \frac{N(N+2)}{4}. \end{aligned}$$

Thus each fixed- N sector is one spin- $j = N/2$ irrep, with dimension $N + 1 = 2j + 1$:

$$n_+ = j + m, \quad n_- = j - m, \quad |jm\rangle = \frac{(a_+^\dagger)^{j+m}(a_-^\dagger)^{j-m}}{\sqrt{(j+m)!(j-m)!}}|0,0\rangle,$$

$$J_+|n_+, n_-\rangle = \hbar\sqrt{(n_+ + 1)n_-}|n_+ + 1, n_- - 1\rangle = \hbar\sqrt{(j - m)(j + m + 1)}|j, m + 1\rangle.$$

Nonnegative integer occupations enforce $2j \in \mathbb{Z}_{\geq 0}$ and $m = -j, \dots, j$; the full Fock space is the direct sum of all these sectors, not one fixed spin. The oscillator representation is kinematic: it does not assert a physical oscillator Hamiltonian for every spin.

4.4.1 Spin-1/2 and the occupation constraint

Source: lecture p. 86; PDF p. 96.

For spin 1/2, restrict *states* to $N = 1$:

$$\mathcal{H}_{1/2} = \text{span}\{|1,0\rangle, |0,1\rangle\}, \quad |+\rangle = a_+^\dagger|0,0\rangle, \quad |-\rangle = a_-^\dagger|0,0\rangle.$$

The bilinears J_i preserve this sector; an individual a_σ or $a_\sigma^\dagger$ does not. In particular,

$$(a_+^\dagger)^2|0,0\rangle = \sqrt{2}|2,0\rangle \neq 0, \quad P_1 a_\sigma^\dagger P_1 = P_1 a_\sigma P_1 = 0,$$

where P_1 projects onto $N = 1$. Consequently, projecting the individual factors before multiplying is *not* equivalent to projecting a spin bilinear: $P_1 a_+^\dagger a_- P_1 \neq (P_1 a_+^\dagger P_1)(P_1 a_- P_1)$. **Hard-core versus constraint:** imposing $(b_\sigma^\dagger)^2 = 0$ changes the on-mode algebra (a hard-core mode has $[b, b^\dagger] = 1 - 2n$, not 1). Even two hard-core modes still admit the empty and doubly occupied states unless $n_+ + n_- = 1$ is imposed. Schwinger bosons remain canonical on the full Fock space; only the physical sector is restricted.

Fermionic alternative: with $\{f_\sigma, f_{\sigma'}^\dagger\} = \delta_{\sigma\sigma'}$, write $\mathbf{S} = (\hbar/2) \sum_{\sigma\sigma'} f_\sigma^\dagger \boldsymbol{\sigma}_{\sigma\sigma'} f_{\sigma'}$ and impose $n_+ + n_- = 1$. The empty and filled sectors are unphysical spin-zero sectors; a site with two fermion modes cannot realize arbitrary j as the bosonic construction can. Auxiliary statistics do not determine the exchange statistics of the actual spin-carrying particles. **Check:** is $|2,0\rangle$ zero or forbidden? **Answer:** nonzero in the auxiliary Fock space, forbidden only for the physical spin-1/2 sector.

4.4.2 Two spin-1/2 systems: triplet and singlet

Source: lecture p. 87; PDF p. 97.

Setting: two independent spin factors, $[S_{1i}, S_{2j}] = 0$, with four product states $|++\rangle, |+-\rangle, |-+\rangle, |--\rangle$ and $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$. If using Schwinger modes, each factor has its *own* $N_i = 1$ constraint; replacing both by one $N = 2$ two-mode sector would retain only the triplet and lose the singlet.

$$\begin{aligned} |1,1\rangle &= |++\rangle, & S_-|++\rangle &= \hbar(|+-\rangle + |-+\rangle) = \hbar\sqrt{2}|1,0\rangle, \\ S_-|1,0\rangle &= \hbar\sqrt{2}|--\rangle, & |1,-1\rangle &= |--\rangle, \\ |1,0\rangle &= \frac{|+-\rangle + |-+\rangle}{\sqrt{2}}, & |0,0\rangle &= \frac{|+-\rangle - |-+\rangle}{\sqrt{2}}. \end{aligned}$$

Orthogonality in the $m = 0$ sector fixes the remaining state; explicitly $S_z|0,0\rangle = S_\pm|0,0\rangle = 0$, hence $\mathbf{S}^2|0,0\rangle = 0$. The triplet has $\mathbf{S}^2 = 2\hbar^2$ and is symmetric under swapping the spin factors; the singlet is antisymmetric.

$$\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{\mathbf{S}^2 - \mathbf{S}_1^2 - \mathbf{S}_2^2}{2} \implies \begin{cases} \hbar^2/4 & (s = 1), \\ -3\hbar^2/4 & (s = 0). \end{cases}$$

Since every total generator annihilates the singlet,

$$(U \otimes U)|0,0\rangle = e^{-i\theta \mathbf{n} \cdot (\mathbf{S}_1 + \mathbf{S}_2)/\hbar}|0,0\rangle = |0,0\rangle.$$

This requires a *common* rotation; $U_1 \otimes U_2$ with unequal rotations generally changes it. Spin-factor symmetry alone is not the full identical-particle exchange rule: orbital and other degrees of freedom also matter.

Qubits and entanglement: $|\psi\rangle = \alpha|++\rangle + \beta|--\rangle$, $|\alpha|^2 + |\beta|^2 = 1$. For a normalized pure two-qubit state $\sum_{\sigma\tau} c_{\sigma\tau}|\sigma\tau\rangle$, factorization requires and is equivalent to $\det(c_{\sigma\tau}) = 0$ (coefficient matrix rank one). For $|\Phi^+\rangle = (|++\rangle + |--\rangle)/\sqrt{2}$, the determinant is $1/2 \neq 0$; for the singlet it is also $1/2$. Both are entangled. The triplet extremes are products, whereas $|1,0\rangle$ is entangled: being in the triplet does not decide separability. **Check:** why is the singlet invariant but $|\Phi^+\rangle$ not invariant under every common rotation? **Answer:** only the singlet is annihilated by all total-spin generators; $|\Phi^+\rangle$ lies in the spin-one sector.

4.4.3 Closed expression for the rotation matrices

Source: lecture p. 88; PDF p. 98.

Let $U_y(\beta) = e^{-i\beta J_y/\hbar}$, $c = \cos(\beta/2)$, $s = \sin(\beta/2)$. Derive the transformed creators rather than guessing rotation signs:

$$\begin{aligned} [J_y, a_+^\dagger] &= \frac{i\hbar}{2} a_-^\dagger, & [J_y, a_-^\dagger] &= -\frac{i\hbar}{2} a_+^\dagger, \\ f_\pm &= U_y a_\pm^\dagger U_y^{-1}, & \frac{df_+}{d\beta} &= \frac{1}{2} f_-, & \frac{df_-}{d\beta} &= -\frac{1}{2} f_+, & f_\pm(0) &= a_\pm^\dagger, \\ \boxed{f_+} &= ca_+^\dagger + sa_-^\dagger, & \boxed{f_-} &= -sa_+^\dagger + ca_-^\dagger. \end{aligned}$$

The vacuum is invariant, so

$$U_y |jm\rangle = \frac{(ca_+^\dagger + sa_-^\dagger)^{j+m} (-sa_+^\dagger + ca_-^\dagger)^{j-m}}{\sqrt{(j+m)!(j-m)!}} |0,0\rangle = \sum_{m'=-j}^j d_{m'm}^{(j)}(\beta) |jm'\rangle.$$

Binomial reconstruction: choose k factors $sa_-^\dagger$ from the first power and r factors $-sa_+^\dagger$ from the second. The final plus occupation gives

$$j + m - k + r = j + m' \implies r = k - m + m'.$$

The coefficient of the unnormalized creator monomial is

$$\binom{j+m}{k} \binom{j-m}{k-m+m'} (-1)^{k-m+m'} c^{2j-2k+m-m'} s^{2k-m+m'}.$$

Multiplying by the final Fock normalization gives the finite sum

$$\begin{aligned} d_{m'm}^{(j)}(\beta) &= \sqrt{(j+m)!(j-m)!(j+m')!(j-m')!} \\ &\times \sum_{k=k_{\min}}^{k_{\max}} \frac{(-1)^{k-m+m'} c^{2j-2k+m-m'} s^{2k-m+m'}}{(j+m-k)! k! (j-m'-k)! (k-m+m')!}, \\ k_{\min} &= \max(0, m-m'), & k_{\max} &= \min(j+m, j-m'). \end{aligned}$$

All factorial arguments and exponents are nonnegative integers for allowed j, m, m' ; the sign exponent is an integer even for half-integer spin. Sum over integer k . At $c = 0$ or $s = 0$, interpret a zero power as 1, as in the underlying polynomial.

Mandatory active-convention sign checks:

$$\begin{aligned} d^{(1/2)}(\beta) &= \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, & d^{(1)}(\beta) &= \begin{pmatrix} c^2 & -\sqrt{2}cs & s^2 \\ \sqrt{2}cs & c^2 - s^2 & -\sqrt{2}cs \\ s^2 & \sqrt{2}cs & c^2 \end{pmatrix}, \\ \partial_\beta d^{(1/2)} \Big|_0 &= \begin{pmatrix} 0 & -1/2 \\ 1/2 & 0 \end{pmatrix} = -\frac{iJ_y}{\hbar}. \end{aligned}$$

In particular $d_{+,-}^{(1/2)} = -s$, not $+s$. For the opposite entry $m = +1/2$, $m' = -1/2$, the only allowed term is $k = 1$: its sign exponent is $k - m + m' = 1 - 1/2 - 1/2 = 0$, so $d_{-,+}^{(1/2)} = +s$. For $m = -1/2$, $m' = +1/2$, only $k = 0$ contributes and the exponent is 1, giving $-s$. Thus the printed sign in Eq. (4.113) is consistent with Eq. (4.85). Exchanging row and column or using U_y^{-1} reverses these off-diagonal signs. Useful global checks are

$$d(0) = 1, \quad d(\beta)^T d(\beta) = 1, \quad d(-\beta) = d(\beta)^T, \quad d^{(j)}(2\pi) = (-1)^{2j} 1.$$

With Euler order $U = e^{-i\alpha J_z/\hbar} U_y(\beta) e^{-i\gamma J_z/\hbar}$, $D_{m'm}^{(j)}(\alpha, \beta, \gamma) = e^{-im'\alpha} d_{m'm}^{(j)}(\beta) e^{-im\gamma}$; the rightmost rotation acts first. **Verification:** direct numerical comparison of this sum with the active matrix exponential for $j = 0, 1/2, 1, 3/2, 2, 3$, at six positive, negative and endpoint angles, agreed to below 10^{-14} . Source Eq. (4.113) has the correct signs in this convention; no correction is needed. **Check:** rotate $|+\rangle$ about y through β . **Answer:** $c|+\rangle + s|-\rangle$, with $\langle \mathbf{J} \rangle = (\hbar/2)(\sin \beta, 0, \cos \beta)$.

4.5 Addition of angular momenta and Clebsch–Gordan coefficients

Source: lecture pp. 88–89; PDF pp. 98–99.

Assumption and bases: $\mathcal{H} = \mathcal{H}_{j_1} \otimes \mathcal{H}_{j_2}$, $[J_{1i}, J_{2k}] = 0$, $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$. Thus $\mathbf{J}$ obeys the angular-momentum algebra. Fix j_1, j_2 ; retain extra spectator labels if the physical Hilbert space has other degeneracies.

Basis	Ket	Diagonal observables
Uncoupled	$ j_1 m_1; j_2 m_2\rangle$	$\mathbf{J}_1^2, \mathbf{J}_2^2, J_{1z}, J_{2z}$
Coupled	$ (j_1 j_2) j m\rangle$	$\mathbf{J}_1^2, \mathbf{J}_2^2, \mathbf{J}^2, J_z$

Since $J_z = J_{1z} + J_{2z}$, only $m_1 + m_2 = m$ appears in a coupled state; individual m_1, m_2 are generally not sharp there. The interaction term is

$$\mathbf{J}^2 = \mathbf{J}_1^2 + \mathbf{J}_2^2 + 2\mathbf{J}_1 \cdot \mathbf{J}_2.$$

Triangle rule by weight counting: suppose $j_1 \geq j_2$. The largest weight $m = j_1 + j_2$ has one product state, yielding one $j = j_1 + j_2$ multiplet. Remove its one state in each weight sector. Before subtraction the number of product states at nonnegative m is

$$n(m) = \begin{cases} 2j_2 + 1, & 0 \leq m \leq j_1 - j_2, \\ j_1 + j_2 - m + 1, & j_1 - j_2 \leq m \leq j_1 + j_2, \end{cases}$$

with m on the allowed integer or half-integer lattice. The multiplicity of spin j is $n(j) - n(j+1)$, namely one for $j_1 - j_2 \leq j \leq j_1 + j_2$ and zero below. Interchanging the labels gives

$$j = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2, \quad \mathcal{H}_{j_1} \otimes \mathcal{H}_{j_2} = \bigoplus_j \mathcal{H}_j.$$

For $j_1 \geq j_2$, there are $2j_2 + 1$ terms with average dimension $2j_1 + 1$, hence $\sum_j (2j+1) = (2j_1+1)(2j_2+1)$. **Check:** can two half-integer spins couple to half-integer j ? **Answer:** no; their sum and difference are integers, and all allowed j are integers.

4.5.1 Clebsch–Gordan coefficients

Source: lecture pp. 89–90; PDF pp. 99–100.

Define the unitary change of basis (fixed j_1, j_2):

$$|(j_1 j_2) j m\rangle = \sum_{m_1, m_2} C_{j_1 m_1, j_2 m_2}^{j m} |j_1 m_1; j_2 m_2\rangle, \\ C_{j_1 m_1, j_2 m_2}^{j m} = \langle j_1 m_1; j_2 m_2 | (j_1 j_2) j m \rangle, \quad C_{j_1 m_1, j_2 m_2}^{j m} = 0 \quad (m_1 + m_2 \neq m).$$

Unitarity, including conjugates before imposing real phases:

$$\sum_{j m} C_{j_1 m_1, j_2 m_2}^{j m} (C_{j_1 m'_1, j_2 m'_2}^{j m})^* = \delta_{m_1 m'_1} \delta_{m_2 m'_2}, \\ \sum_{m_1 m_2} (C_{j_1 m_1, j_2 m_2}^{j m})^* C_{j_1 m'_1, j_2 m'_2}^{j' m'} = \delta_{j j'} \delta_{m m'}, \\ |j_1 m_1; j_2 m_2\rangle = \sum_{j m} (C_{j_1 m_1, j_2 m_2}^{j m})^* |(j_1 j_2) j m\rangle.$$

The standard real Condon–Shortley convention uses positive ladder coefficients and fixes, for each j , the highest-weight coefficient with largest allowed m_1 to be positive (here $m_1 = j_1$, $m_2 = j - j_1$). Reality alone does not fix the remaining overall multiplet sign.

Recursion derivation: apply $J_+ = J_{1+} + J_{2+}$ to the expansion of $|j m\rangle$ and compare the coefficient of a product state with $m_1 + m_2 = m + 1$:

$$\begin{aligned} & \sqrt{(j-m)(j+m+1)} C_{j_1 m_1, j_2 m_2}^{j, m+1} \\ &= \sqrt{(j_1+m_1)(j_1-m_1+1)} C_{j_1, m_1-1; j_2 m_2}^{j m} \\ & \quad + \sqrt{(j_2+m_2)(j_2-m_2+1)} C_{j_1 m_1; j_2, m_2-1}^{j m}. \end{aligned}$$

Equivalently, lowering and comparing a state with $m_1 + m_2 = m - 1$ gives

$$\begin{aligned} & \sqrt{(j+m)(j-m+1)} C_{j_1 m_1, j_2 m_2}^{j, m-1} \\ &= \sqrt{(j_1-m_1)(j_1+m_1+1)} C_{j_1, m_1+1; j_2 m_2}^{j m} \\ & \quad + \sqrt{(j_2-m_2)(j_2+m_2+1)} C_{j_1 m_1; j_2, m_2+1}^{j m}. \end{aligned}$$

Out-of-range coefficients vanish. The factors on the right refer to the *input* product weights, not the output weights; this distinction prevents off-by-one errors. The common $\hbar$ cancels.

Highest-weight algorithm, sufficient to reconstruct every coefficient:

1. Begin with $|j_1 + j_2, j_1 + j_2\rangle = |j_1, j_1; j_2, j_2\rangle$.
2. For a chosen j , find a normalized vector in the $m = j$ sector orthogonal to all already constructed larger- j states in that sector. Equivalently solve $J_+|j, j\rangle = 0$. Fix its overall phase as stated above.
3. Generate the entire multiplet using $|j, m-1\rangle = J_-|j, m\rangle / [\hbar\sqrt{(j+m)(j-m+1)}]$.
4. Decrease j by one and repeat down to $|j_1 - j_2|$; verify norms, mutual orthogonality, selection rules and the total dimension.

For an explicit highest-weight linear equation, set $c_{m_1, m_2} = C_{j_1 m_1, j_2 m_2}^{jj}$ on $m_1 + m_2 = j$. The coefficient of $|j_1 \mu_1; j_2 \mu_2\rangle$ with $\mu_1 + \mu_2 = j + 1$ must obey

$$\sqrt{(j_1 + \mu_1)(j_1 - \mu_1 + 1)}c_{\mu_1-1, \mu_2} + \sqrt{(j_2 + \mu_2)(j_2 - \mu_2 + 1)}c_{\mu_1, \mu_2-1} = 0.$$

Do not choose the phase independently at every m : once the highest-weight phase is fixed, lowering fixes all relative phases within that multiplet. **Check:** does orthogonality alone determine the sign of a new multiplet? **Answer:** no; normalization and orthogonality leave one overall phase, which the phase convention fixes.

4.5.2 Example: coupling $l = 1$ and $s = 1/2$

Source: lecture pp. 90–91; PDF pp. 100–101.

Abbreviate $|m_l, \pm\rangle = |1, m_l\rangle \otimes |1/2, \pm 1/2\rangle$ and $|j, m\rangle = |(1, 1/2)jm\rangle$. The six-dimensional product decomposes as $j = 3/2 \oplus 1/2$ (dimensions $4 + 2$).

$$\begin{aligned} |\tfrac{3}{2}, \tfrac{3}{2}\rangle &= |1, +\rangle, \\ J_-|1, +\rangle &= \hbar(\sqrt{2}|0, +\rangle + |1, -\rangle) = \hbar\sqrt{3}|\tfrac{3}{2}, \tfrac{1}{2}\rangle. \end{aligned}$$

For the second highest weight write $|\tfrac{1}{2}, \tfrac{1}{2}\rangle = A|1, -\rangle + B|0, +\rangle$. Then

$$J_+|\tfrac{1}{2}, \tfrac{1}{2}\rangle = \hbar(A + \sqrt{2}B)|1, +\rangle = 0, \quad |A|^2 + |B|^2 = 1, \quad A > 0,$$

so $A = \sqrt{2/3}$, $B = -\sqrt{1/3}$. Lowering once gives

$$J_-|\tfrac{1}{2}, \tfrac{1}{2}\rangle = \hbar\left(\sqrt{\tfrac{1}{3}}|0, -\rangle - \sqrt{\tfrac{2}{3}}|-1, +\rangle\right) = \hbar|\tfrac{1}{2}, -\tfrac{1}{2}\rangle.$$

Complete CG table (standard Condon–Shortley convention):

(j, m)	Coupled state in the product basis
$(\tfrac{3}{2}, \tfrac{3}{2})$	$ 1, +\rangle$
$(\tfrac{3}{2}, \tfrac{1}{2})$	$\sqrt{\tfrac{1}{3}} 1, -\rangle + \sqrt{\tfrac{2}{3}} 0, +\rangle$
$(\tfrac{3}{2}, -\tfrac{1}{2})$	$\sqrt{\tfrac{2}{3}} 0, -\rangle + \sqrt{\tfrac{1}{3}} -1, +\rangle$
$(\tfrac{3}{2}, -\tfrac{3}{2})$	$ -1, -\rangle$
$(\tfrac{1}{2}, \tfrac{1}{2})$	$\sqrt{\tfrac{2}{3}} 1, -\rangle - \sqrt{\tfrac{1}{3}} 0, +\rangle$
$(\tfrac{1}{2}, -\tfrac{1}{2})$	$\sqrt{\tfrac{1}{3}} 0, -\rangle - \sqrt{\tfrac{2}{3}} -1, +\rangle$

Spin–orbit application: these states diagonalize $\mathbf{L} \cdot \mathbf{S}$, since

$$\mathbf{L} \cdot \mathbf{S} = \frac{\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2}{2} \implies \begin{cases} \hbar^2/2, & j = 3/2, \\ -\hbar^2, & j = 1/2. \end{cases}$$

For $H = \lambda \mathbf{L} \cdot \mathbf{S}$, multiply these eigenvalues by λ (do not drop $\hbar^2$ unless it is absorbed into the definition of λ). For example, measuring J^2 in $|0, +\rangle$ gives probabilities $2/3$ for $j = 3/2$ and $1/3$ for $j = 1/2$; amplitudes depend on phase convention, probabilities do not.

Source audit and convention pitfall. Source Eqs. (4.134)–(4.135) give *both* $j = 1/2$ states with the opposite overall sign from the table above. They are mutually consistent under lowering and physically equivalent, not a relative-sign error. However, taken with the preceding claim of the standard Condon–Shortley convention, they use a different overall doublet phase. The table here fixes the largest- m_l highest-weight coefficient positive and matches the standard CG convention. Source Eq. (4.113) agrees with the active d convention; the extracted spin-one matrix on p. 85 is typographically scrambled, so use the explicitly reconstructed matrix above rather than infer an error from extraction alone. No other mathematical error was established in Sections 4.4–4.5. **Verification:** symbolic CG coefficients agree with all six table rows; exact matrix tests give orthonormality and every lowering relation. The source doublet is exactly minus the standard doublet as a whole.

Self-test for 4.4–4.5: reconstruct before reading the key

1. Derive $\mathbf{J}^2$ from J_+J_- and J_-J_+ . Which spin does $N = 3$ realize, and how many states are there?
2. Why are Schwinger bosons not hard-core? Why must two independently represented spin-1/2 systems have two separate constraints?
3. Construct the two $m = 0$ coupled states by lowering and orthogonality. Which is invariant under every common rotation?
4. In the finite d sum, which combinatorial choice produces the minus sign? Determine $d_{+,-}^{(1/2)}$ and $d_{-,+}^{(1/2)}$ without memorizing the formula.
5. What is the range of j for $j_1 = 2, j_2 = 3/2$? Check the product dimension.
6. Derive the raising recursion by coefficient comparison. What output-weight constraint belongs to it?
7. Find the $j = 1/2, m = 1/2$ highest weight for $l = 1, s = 1/2$; lower it and explain why reversing only the lower state's sign is invalid with positive ladders.
8. For $|1, -\rangle$, give the probabilities of measuring $j = 3/2$ and $j = 1/2$, and the two possible $\mathbf{L} \cdot \mathbf{S}$ values.

Compact key. **1.** $\mathbf{J}^2/\hbar^2 = (N_+ - N_-)^2/4 + [N_+(N_- + 1) + N_-(N_+ + 1)]/2 = N(N + 2)/4$; $j = 3/2$, four states. **2.** $(a_\sigma^\dagger)^2 \neq 0$ in the full Fock space; $N_i = 1$ for each factor yields the full four-state space, whereas one $N = 2$ sector gives only spin one. **3.** $(|+-\rangle \pm |-+\rangle)/\sqrt{2}$; the minus state is the invariant singlet. **4.** Each plus creator chosen from $f_- = -sa_+^\dagger + ca_-^\dagger$ contributes -1 ; the entries are $-s$ and $+s$. **5.** $j = 1/2, 3/2, 5/2, 7/2$, with dimensions $2 + 4 + 6 + 8 = 20 = 5 \times 4$. **6.** Expand $J_+|jm\rangle$ and compare a fixed product ket on each side; its weights satisfy $m_1 + m_2 = m + 1$. **7.** $\sqrt{2/3}|1, -\rangle - \sqrt{1/3}|0, +\rangle$; lowering gives $\sqrt{1/3}|0, -\rangle - \sqrt{2/3}|-1, +\rangle$. Only a common phase change of both states preserves their positive ladder relation. **8.** $1/3$ and $2/3$; $\hbar^2/2$ and $-\hbar^2$, respectively.

4.6 Spin correlations, entanglement, and Bell inequalities

Source: lecture pp. 91–94 (PDF pp. 101–104). Use dimensionless outcomes $s, t = \pm 1$: physical spin outcomes are $\hbar s/2$. Tensor factors label distinguishable measurement stations; A and B commute because they act on different factors.

4.6.1 The singlet state and its correlations

In the ordered z basis $(|++\rangle, |+-\rangle, |-+\rangle, |--\rangle)$,

$$|\psi^-\rangle = \frac{|+-\rangle - |-+\rangle}{\sqrt{2}}, \quad (\sigma_1 + \sigma_2)|\psi^-\rangle = 0,$$

$$\rho = |\psi^-\rangle\langle\psi^-| = \frac{1}{4} \left(I \otimes I - \sum_{i=x,y,z} \sigma_i \otimes \sigma_i \right), \quad (U \otimes U)|\psi^-\rangle = |\psi^-\rangle \quad (U \in SU(2)).$$

Reconstruction: $\langle\sigma_{1z}\sigma_{2z}\rangle = -1$ by inspection; rotational invariance fixes $\langle\sigma_{1i}\sigma_{2j}\rangle = -\delta_{ij}$ and $\langle\sigma_{1i}\rangle = \langle\sigma_{2i}\rangle = 0$. For unit axes $\mathbf{a}, \mathbf{b}$, define $A = \sigma_1 \cdot \mathbf{a}$ and $B = \sigma_2 \cdot \mathbf{b}$:

$$E(\mathbf{a}, \mathbf{b}) = \sum_{ij} a_i b_j \langle\sigma_{1i}\sigma_{2j}\rangle = \boxed{-\mathbf{a} \cdot \mathbf{b} = -\cos\theta_{ab}},$$

$$\Pi_s(\mathbf{a}) = \frac{I + s\sigma \cdot \mathbf{a}}{2}, \quad P(s, t|\mathbf{a}, \mathbf{b}) = \text{Tr}[\rho \Pi_s(\mathbf{a}) \otimes \Pi_t(\mathbf{b})] = \boxed{\frac{1 - sta \cdot \mathbf{b}}{4}},$$

$$P(+, +) = P(-, -) = \frac{1 - \cos\theta_{ab}}{4} = \frac{1}{2} \sin^2 \frac{\theta_{ab}}{2}, \quad P(+, -) = P(-, +) = \frac{1 + \cos\theta_{ab}}{4}.$$

Check $\sum_{s,t} P = 1$ and $\sum_{s,t} stP = E$. Equal axes: opposite signs each with probability 1/2; perpendicular axes: all four probabilities 1/4; reversed axes: equal signs each with probability 1/2. Joint probability $P(+, +)$ is *not* the probability of either equal-sign result, which is $2P(+, +)$.

Local state and no signalling:

$$\rho_A = \text{Tr}_B \rho = I/2, \quad \rho_B = \text{Tr}_A \rho = I/2, \quad P_B(t|\mathbf{b}) = \sum_s P(s, t|\mathbf{a}, \mathbf{b}) = 1/2,$$

$$P(t|s, \mathbf{a}, \mathbf{b}) = \frac{1 - sta \cdot \mathbf{b}}{2}, \quad \rho_{B|s, \mathbf{a}} = \frac{I - s\sigma \cdot \mathbf{a}}{2}, \quad \sum_s \frac{1}{2} \rho_{B|s, \mathbf{a}} = I/2.$$

Conditioning on Alice's outcome changes Bob's ensemble description; without that outcome (which requires classical communication), Bob sees the same mixed state. More generally a local trace-preserving operation with

$\sum_k M_k^\dagger M_k = I$ gives $\text{Tr}_A \sum_k (M_k \otimes I) \rho (M_k^\dagger \otimes I) = \rho_B$. Postselection is not trace-preserving and must not be mistaken for a controllable remote signal. A maximally mixed reduced state does not mean the joint state is an uncorrelated mixture: $\rho \neq \rho_A \otimes \rho_B$.

4.6.2 The EPR question and local hidden variables

EPR (1935) used position/momentum; Bohm's spin version replaces them by alternative axes with perfect singlet anticorrelation. A certain inference of Bob's result after Alice's measurement is not an experimental simultaneous assignment of S_x and S_z . The testable local model assumes

$$p(\lambda|\mathbf{a}, \mathbf{b}) = p(\lambda), \quad p(\lambda) \geq 0, \quad \int d\lambda p(\lambda) = 1,$$

$$P(s, t|\mathbf{a}, \mathbf{b}, \lambda) = P_A(s|\mathbf{a}, \lambda) P_B(t|\mathbf{b}, \lambda), \quad E(\mathbf{a}, \mathbf{b}) = \int d\lambda p(\lambda) A(\mathbf{a}, \lambda) B(\mathbf{b}, \lambda).$$

For deterministic models $A, B = \pm 1$; stochastic local models can be made deterministic by enlarging λ with local random seeds, or bounded using local means in $[-1, 1]$. **Assumptions:** locality in the factorization sense, measurement-setting independence from λ , and one common ensemble for the compared settings. No-signalling alone is weaker than this factorization.

Three-axis Bell inequality: with *perfect* anticorrelation, $B(\mathbf{d}, \lambda) = -A(\mathbf{d}, \lambda)$ for each tested direction (up to zero-measure exceptions). If $A(\mathbf{a}) = +$ and $B(\mathbf{b}) = +$, then $A(\mathbf{b}) = -$; either $A(\mathbf{c}) = -$, giving $(a+, c+)$, or $A(\mathbf{c}) = +$, giving $(c+, b+)$. Thus the event inclusion gives

$$P(a+; b+) \leq P(a+; c+) + P(c+; b+),$$

$$\theta_{ac} = \theta_{cb} = \theta, \quad \theta_{ab} = 2\theta \implies \sin^2 \theta \leq 2 \sin^2(\theta/2).$$

For $0 < \theta < \pi/2$ the difference is $2 \sin^2(\theta/2) \cos \theta > 0$, so quantum mechanics violates the bound. At $\theta = 60^\circ$, the *scaled* sides are 3/4 and 1/2; the actual joint-probability sides are 3/8 and 1/4. **Validity pitfall:** the three-setting argument uses perfect anticorrelation, unlike the CHSH bound below.

4.6.3 The CHSH inequality

For two settings per station, let $E_{ij} = E(\mathbf{a}_i, \mathbf{b}_j)$ and $S = |E_{11} + E_{12} + E_{21} - E_{22}|$. With deterministic local outcomes,

$$C(\lambda) = A_1(B_1 + B_2) + A_2(B_1 - B_2) = \pm 2,$$

$$B_1 = B_2 \Rightarrow B_1 - B_2 = 0, \quad B_1 = -B_2 \Rightarrow B_1 + B_2 = 0,$$

$$\boxed{S = \left| \int d\lambda p(\lambda) C(\lambda) \right| \leq \int d\lambda p(\lambda) |C(\lambda)| = 2.}$$

For stochastic local means, $|C| \leq |B_1 + B_2| + |B_1 - B_2| \leq 2$ gives the same result. The common $p(\lambda)$ is essential; selected detection samples require justification or a loophole-free protocol. For a singlet choose

$$\mathbf{a}_1 = \hat{\mathbf{z}}, \quad \mathbf{a}_2 = \hat{\mathbf{x}}, \quad \mathbf{b}_1 = \frac{\hat{\mathbf{z}} + \hat{\mathbf{x}}}{\sqrt{2}}, \quad \mathbf{b}_2 = \frac{\hat{\mathbf{z}} - \hat{\mathbf{x}}}{\sqrt{2}},$$

$$(E_{11}, E_{12}, E_{21}, E_{22}) = \frac{1}{\sqrt{2}}(-1, -1, -1, +1), \quad \boxed{S = 2\sqrt{2}}.$$

Tsirelson bound, not just one example: for Hermitian observables $A_i^2 = B_j^2 = I$ on separate factors, introduce

$$\mathcal{C} = A_1 \otimes (B_1 + B_2) + A_2 \otimes (B_1 - B_2),$$

$$\mathcal{C}^2 = 4I - [A_1, A_2] \otimes [B_1, B_2], \quad \|[A_1, A_2]\|, \|[B_1, B_2]\| \leq 2,$$

$$|\langle \mathcal{C} \rangle| \leq \|\mathcal{C}\| \leq \sqrt{4 + 4} = 2\sqrt{2}.$$

The displayed orthogonal pairs attain the bound. Commuting settings on either side give $\mathcal{C}^2 = 4I$, hence no CHSH violation. Bell violation excludes the conjunction of the local-model assumptions, not all hidden-variable theories or a unique interpretation; it supplies no controllable faster-than-light communication. For linear photon polarization the analogous correlations involve $2(\alpha - \beta)$, not the physical spin-axis angle; signs depend on which Bell state and outcome labeling are used.

4.7 From Bell tests to the 2022 Nobel Prize in Physics

Source: lecture pp. 94–98 (PDF pp. 104–108). Historical statements below are compressed from the lecture, not new historical claims. The 2022 prize jointly recognized Alain Aspect, John F. Clauser and Anton Zeilinger for entangled-photon experiments, Bell-inequality violations and pioneering quantum-information science.

4.7.1 The Freedman–Clauser experiment: the first decisive Bell test

Bell (1964) → experimental CHSH formulation (1969) → Freedman–Clauser calcium-cascade photon test at Berkeley (1972). Oppositely directed correlated photons pass polarization analyzers; record coincidence rates versus angle, control accidental coincidences and collection losses. The lecture reports improved pile-of-plates polarizers, about 5 m separation and about 200 hours of acquisition, with

$$\delta = \frac{R(22.5^\circ) - R(67.5^\circ)}{R_0} - \frac{1}{4} \leq 0 \quad (\text{relevant local models}), \quad \delta_{\text{exp}} = 0.050 \pm 0.008.$$

R denotes measured coincidence rates; R_0 is the source’s reference-rate normalization, whose detailed measurement prescription is not specified in this extract. The measured positive value violated the experimental inequality and agreed with the predicted quantum curve. This rate inequality is not the ideal four-correlator CHSH formula; inefficient detections introduce extra sampling/experimental assumptions, so do not call this early test loophole-free.

4.7.2 Aspect’s experiments: changing the settings during flight

The 1981–1982 program increased useful calcium-pair rates using laser excitation, then used two-channel polarizers. In 1982 fast acousto-optical switches routed photons between alternative analyzer settings in flight: switching about 10 ns, station separation about 12 m. **Purpose:** prevent a setting from being communicated subluminally to the distant station before detection. The source explicitly notes limited detection efficiency and not-yet-fully-random setting choices; this addressed locality much more strongly but was not the final loophole-free test.

4.7.3 Zeilinger: entanglement as an experimental resource

The source highlights a 1998 Bell experiment with roughly 400 m observer separation, a 1997 teleportation experiment, and 1998 entanglement swapping by Pan, Bouwmeester, Weinfurter and Zeilinger; also multiphoton GHZ states, cryptography and later loophole-free tests. **Swapping reconstruction:** define

$$|\Phi^\pm\rangle = (|00\rangle \pm |11\rangle)/\sqrt{2}, \quad |\Psi^\pm\rangle = (|01\rangle \pm |10\rangle)/\sqrt{2}.$$

With ordered pairs (1, 4) and (2, 3) and the underlying tensor order (1, 2, 3, 4),

$$|\Phi^+\rangle_{12}|\Phi^+\rangle_{34} = \frac{1}{2}(|\Phi^+\rangle_{14}|\Phi^+\rangle_{23} + |\Phi^-\rangle_{14}|\Phi^-\rangle_{23} + |\Psi^+\rangle_{14}|\Psi^+\rangle_{23} + |\Psi^-\rangle_{14}|\Psi^-\rangle_{23}).$$

A Bell outcome on 23 occurs with probability 1/4 and conditions 14 into the corresponding Bell state, despite no direct 1–4 interaction. Without the outcome record, $\rho_{14} = \frac{1}{4}\sum_\nu |\nu\rangle\langle\nu| = I_4/4$: conditioning and classical communication remain essential. **Phase/order check:** the all-plus expansion in source Eq. (4.152) is correct for (14), (23); swapping an ordered singlet’s labels introduces a minus sign. Do not import a differently ordered swapping formula silently. This conditioning mechanism motivates repeaters and networks; the algebra does not itself assert that a particular experiment resolves every Bell outcome.

4.7.4 What the experiments establish—and what they do not

Observed violations agree with quantum mechanics and exclude local hidden-variable models satisfying the tested assumptions; the lecture states that modern experiments close major locality and detection loopholes simultaneously. They neither permit superluminal signalling nor choose a unique interpretation. Distinguish the loopholes: *locality* concerns causal separation of choices/detections; *detection* concerns unrepresentative selected pairs; *measurement independence* concerns setting– λ correlations. Closing major experimental loopholes does not turn the conclusion into one free of stated assumptions. Selected source references: Bell, *Physics* **1**, 195 (1964); CHSH, *PRL* **23**, 880 (1969); Freedman–Clauser, *PRL* **28**, 938 (1972); Aspect et al., *PRL* **47**, 460 (1981), **49**, 91 and 1804 (1982); Weihs et al., *PRL* **81**, 5039 (1998); Bouwmeester et al., *Nature* **390**, 575 (1997); Pan et al., *PRL* **80**, 3891 (1998); Nobel Committee scientific background (2022).

4.8 Problems: solution roadmaps and final checks

Source: lecture pp. 98–99 (PDF pp. 108–109); lecture p. 100 (PDF p. 110) has no extracted content. Throughout use $|z+\rangle = (1, 0)^T$, $|z-\rangle = (0, 1)^T$, right-handed axes, active rotations $U = e^{-i\phi \mathbf{n} \cdot \mathbf{J}/\hbar}$, and standard positive ladder coefficients.

Problem 1: σ_y eigenstates and measurement probability

Solve $(\sigma_y - sI)v = 0$ and normalize:

$$s = \pm 1, \quad |y\pm\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}, \quad \boxed{P(S_y = +\hbar/2) = \frac{|\alpha - i\beta|^2}{2} = \frac{1}{2} + \text{Im}(\alpha^* \beta)}.$$

Assume $|\alpha|^2 + |\beta|^2 = 1$. **Check:** $\sigma_y|y\pm\rangle = \pm|y\pm\rangle$; probabilities sum to one; $\beta = i\alpha$ with $\alpha = 1/\sqrt{2}$ gives certainty for $y+$, not $y-$.

Problem 2: transverse angular-momentum moments

$L_{\pm}|lm\rangle = \hbar\sqrt{l(l+1) - m(m\pm 1)}|l, m\pm 1\rangle$, so diagonal ladder matrix elements and those of $L_{\pm}^2$ vanish.

$$L_x = (L_+ + L_-)/2, \quad L_y = (L_+ - L_-)/(2i), \quad L_+L_- + L_-L_+ = 2(L^2 - L_z^2),$$

$$\boxed{\langle L_x \rangle = \langle L_y \rangle = 0, \quad \langle L_x^2 \rangle = \langle L_y^2 \rangle = \frac{\hbar^2}{2}[l(l+1) - m^2].}$$

Interpretation: fixed length $\hbar\sqrt{l(l+1)}$ and fixed z projection $\hbar m$, with no preferred azimuth; transverse mean vanishes, transverse spread does not. This cone picture is semiclassical, not simultaneous sharp Cartesian components. **Check:** sum of three second moments is $\hbar^2 l(l+1)$; even $m = l > 0$ retains $\langle L_x^2 \rangle = \hbar^2 l/2$.

Problem 3: arbitrary-axis L_n , outcomes and moments

For $\mathbf{n} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta)$,

$$L_n = \frac{\sin \theta}{2}(e^{-i\varphi}L_+ + e^{i\varphi}L_-) + L_z \cos \theta.$$

(a) **Spectrum/outcomes:** $R = e^{-i\varphi L_z/\hbar} e^{-i\theta L_y/\hbar}$ gives $L_n = RL_zR^\dagger$, so eigenvalues are $\hbar\mu$, $\mu = -l, \dots, l$, with probabilities $|\langle l\mu|R^\dagger|lm\rangle|^2$. An eigenvalue is an actual possible outcome only if its probability is nonzero (for $\theta = 0$, only $\mu = m$ occurs). (b) **Moments:** $\langle \{L_i, L_j\} \rangle = 0$ for $i \neq j$; use Problem 2:

$$\boxed{\langle L_n \rangle = \hbar m \cos \theta, \quad \langle L_n^2 \rangle = \hbar^2 \left[m^2 \cos^2 \theta + \frac{1}{2}(l(l+1) - m^2) \sin^2 \theta \right],}$$

$$(\Delta L_n)^2 = \frac{\hbar^2}{2}[l(l+1) - m^2] \sin^2 \theta.$$

Check: $\theta = 0$ gives zero variance; $\theta = \pi/2$ recovers transverse moments; φ drops out by axial symmetry.

Problem 4: explicit spin-1/2 rotation

Distinguish the rotation angle ϕ from the axis azimuth φ . Since $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$, sum even and odd powers:

$$U = \cos(\phi/2)I - i \sin(\phi/2)\mathbf{n} \cdot \boldsymbol{\sigma},$$

$$\boxed{U = \begin{pmatrix} c - is \cos \theta & -is \sin \theta e^{-i\varphi} \\ -is \sin \theta e^{i\varphi} & c + is \cos \theta \end{pmatrix}}, \quad c = \cos(\phi/2), \quad s = \sin(\phi/2).$$

$U^\dagger U = (c^2 + s^2)I = I$, $\det U = 1$, $U(2\pi) = -I$, $U(4\pi) = I$. **Check:** $U(\hat{\mathbf{z}}, \phi) = \text{diag}(e^{-i\phi/2}, e^{i\phi/2})$. A 2π sign is globally unobservable for an isolated spinor, but a relative sign can matter in interference.

Problem 5: Euler matrix and axis-angle representation

Multiply the z - y - z factors, with the rightmost acting first:

$$\boxed{D^{(1/2)}(\alpha, \beta, \gamma) = \begin{pmatrix} e^{-i(\alpha+\gamma)/2} \cos(\beta/2) & -e^{-i(\alpha-\gamma)/2} \sin(\beta/2) \\ e^{i(\alpha-\gamma)/2} \sin(\beta/2) & e^{i(\alpha+\gamma)/2} \cos(\beta/2) \end{pmatrix}}.$$

Every $SU(2)$ matrix has $D = q_0 I - i\mathbf{q} \cdot \boldsymbol{\sigma}$, with real $q_0, \mathbf{q}$ and $q_0^2 + |\mathbf{q}|^2 = 1$.

$$\boxed{\cos(\phi/2) = q_0 = \frac{1}{2} \text{Tr } D = \cos(\beta/2) \cos[(\alpha + \gamma)/2]}, \quad q_i = \frac{i}{2} \text{Tr}(\sigma_i D),$$

$$\phi = 2 \arccos q_0 \in [0, 2\pi], \quad \mathbf{n} = \mathbf{q} / \sin(\phi/2) \quad \text{if } \sin(\phi/2) \neq 0.$$

Hence $D = e^{-i(\phi/2)\mathbf{n} \cdot \boldsymbol{\sigma}}$. **Check:** $\beta = 0$ reduces to a z rotation by $\alpha + \gamma$; $D = \pm I$ has no uniquely determined axis. Trace alone determines the half-angle cosine, not the axis or all branch choices.

Problem 6: two-spin Clebsch–Gordan coefficients

Start from the highest weight and lower with $J_- = S_{1-} + S_{2-}$; normalize, then choose the orthogonal antisymmetric state:

$$|1, 1\rangle = |++\rangle, \quad |1, 0\rangle = (|+-\rangle + |-+\rangle)/\sqrt{2}, \quad |1, -1\rangle = |--\rangle,$$

$$|0, 0\rangle = (|+-\rangle - |-+\rangle)/\sqrt{2}, \quad \mathbf{J} = \mathbf{S}_1 + \mathbf{S}_2, \quad J^2 = \frac{3}{2}\hbar^2 I + \frac{1}{2}\hbar^2 \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2.$$

These coefficients are the required CG coefficients; omitted coefficients are zero. $\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$ has eigenvalue $+1$ on the triplet and -3 on the singlet, giving $J^2 = 2\hbar^2$ and 0 , respectively; $J_z = \hbar m$. **Check:** four orthonormal states span the product space; singlet is annihilated by every J_i ; a whole multiplet's overall phase is conventional.

Problem 7: Cartesian generators and the $j = 1$ basis

Write $G_i = \hbar g_i$ where

$$g_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad g_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad g_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Multiply these matrices (or contract Levi-Civita symbols): $[G_i, G_j] = i\hbar \epsilon_{ijk} G_k$. Choose the columns of the Cartesian-to-spherical basis matrix as $\mathbf{e}_+ = -(\hat{\mathbf{x}} + i\hat{\mathbf{y}})/\sqrt{2}$, $\mathbf{e}_0 = \hat{\mathbf{z}}$, $\mathbf{e}_- = (\hat{\mathbf{x}} - i\hat{\mathbf{y}})/\sqrt{2}$:

$$C = \begin{pmatrix} -1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -i/\sqrt{2} & 0 & -i/\sqrt{2} \\ 0 & 1 & 0 \end{pmatrix}, \quad C^\dagger C = I, \quad \mathbf{V}_{\text{cart}} = C \mathbf{V}_{\text{sph}},$$

$$C^\dagger G_z C = \hbar \text{diag}(1, 0, -1), \quad C^\dagger G_x C = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$C^\dagger G_y C = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

Infinitesimal active rotation: $\delta V_j = -(i/\hbar) \delta \phi_i (G_i)_{jk} V_k = -\delta \phi_i \epsilon_{ijk} V_k = (\delta \boldsymbol{\phi} \times \mathbf{V})_j$. **Check:** a small positive z rotation sends $\hat{\mathbf{x}}$ toward $\hat{\mathbf{y}}$; $\sum_i G_i^2 = 2\hbar^2 I$. A passive coordinate rotation uses the opposite sign; confusing C with $C^\dagger$ reverses the basis conversion.

Problem 8: coupling $1 \otimes \frac{1}{2}$ by lowering

Start with $|\frac{3}{2}, \frac{3}{2}\rangle = |1, 1\rangle |\frac{1}{2}, \frac{1}{2}\rangle$.

$$J_- |\frac{3}{2}, \frac{3}{2}\rangle = \hbar \sqrt{3} |\frac{3}{2}, \frac{1}{2}\rangle$$

$$= \hbar |1, 1\rangle |\frac{1}{2}, -\frac{1}{2}\rangle + \hbar \sqrt{2} |1, 0\rangle |\frac{1}{2}, \frac{1}{2}\rangle.$$

Thus $\boxed{a = 1/\sqrt{3}, \quad b = \sqrt{2/3}}$ and one consistent orthogonal choice is

$$|\frac{1}{2}, \frac{1}{2}\rangle = \sqrt{\frac{2}{3}} |1, 1\rangle |\frac{1}{2}, -\frac{1}{2}\rangle - \frac{1}{\sqrt{3}} |1, 0\rangle |\frac{1}{2}, \frac{1}{2}\rangle.$$

Check: both normalized, overlap zero; J^2 eigenvalues $15\hbar^2/4$ and $3\hbar^2/4$; J_+ annihilates the $j = 1/2$ state. Its overall minus is equally acceptable, but changing only one relative sign is not.

Problem 9: singlet tensor and joint probability

Roadmap: evaluate $\langle \sigma_{1i} \sigma_{2j} \rangle = -\delta_{ij}$ directly in $(|+-\rangle - |-+\rangle)/\sqrt{2}$, or use rotational invariance and one diagonal element; contract with $a_i b_j$. Then multiply the $+$ projectors rather than confusing amplitudes with conditional probabilities:

$$\boxed{E = -\mathbf{a} \cdot \mathbf{b}}, \quad P(+, +) = \frac{1}{4} \langle (I + A)(I + B) \rangle = \boxed{\frac{1}{4}(1 - \mathbf{a} \cdot \mathbf{b}) = \frac{1}{2} \sin^2(\theta_{ab}/2)}.$$

Check: zero for equal axes, $1/4$ for perpendicular axes, $1/2$ for opposite axes; the dimensionful correlation is $\langle S_{1a} S_{2b} \rangle = -(\hbar^2/4) \mathbf{a} \cdot \mathbf{b}$.

Problem 10: CHSH violation and Bob's reduced state

Take the four dot products in the stated order:

$$E_{11} = E_{12} = E_{21} = -1/\sqrt{2}, \quad E_{22} = +1/\sqrt{2}, \quad E_{11} + E_{12} + E_{21} - E_{22} = -2\sqrt{2}, \quad \boxed{S = 2\sqrt{2}}.$$

The density matrix is

$$\rho = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (\rho_B)_{bc} = \sum_{a=0,1} \rho_{ab,ac}, \quad \boxed{\rho_B = I/2}.$$

A nonselective Alice measurement yields $\rho'_B = \sum_{s=\pm 1} \frac{1}{2} \rho_{B|s,\mathbf{a}} = I/2$, independent of $\mathbf{a}$. **Check:** each Bob outcome stays $1/2$; correlations require comparing records, so violation does not create a signalling channel.

Self-test and compact key for 4.6–4.8

1. At $\mathbf{a} \cdot \mathbf{b} = 1/2$, find $P(+, +)$, $P_B(+)$ and $P(B+|A+)$. Why do these differ?
2. Which extra assumption distinguishes the three-axis Bell proof from CHSH? What common statistical assumption do both need?
3. Reconstruct the CHSH bound and the quantum operator-square identity. Why can commuting local settings not violate it?
4. After entanglement swapping, what does Alice see before receiving the Bell-outcome record? Does a mixed local state imply a separable joint state?
5. In $|lm\rangle$, what is $(\Delta L_n)^2$? Why is zero transverse mean not zero transverse angular momentum?
6. What does the trace of an $SU(2)$ Euler matrix determine? Where does axis extraction fail?

Key. **1.** $1/8, 1/2, 1/4$: joint, marginal and conditional distributions have different normalization. **2.** Perfect anticorrelation; both need setting independence and a common hidden-variable ensemble with local responses. **3.** One of $B_1 \pm B_2$ vanishes for deterministic outcomes; $\mathcal{C}^2 = 4I - [A_1, A_2] \otimes [B_1, B_2]$; a zero commutator forces norm 2. **4.** Before conditioning, $\rho_{14} = I_4/4$ and each single spin is $I/2$; a mixed reduced state alone does not imply a separable joint state (the singlet is the counterexample). **5.** $\hbar^2[l(l+1) - m^2] \sin^2 \theta/2$; nonzero second moments encode the spread. **6.** $\cos(\phi/2)$, not the axis; $D = \pm I$ leaves the axis undetermined.

Source audit and verification. No definite algebraic source error was found in Sections 4.6–4.8. Source Eq. (4.145)'s 60° example compares probabilities after canceling a common $1/2$, not raw coincidence probabilities. Eq. (4.152)'s all-plus swapping signs were verified with its ordered labels. R_0 in Eq. (4.150) is not operationally defined in the supplied extract, so no unprovided prescription is asserted here. Damaged extraction typography was reconstructed using the displayed definitions; preceding CG equations (4.134)–(4.135) on lecture p. 91 belong to Section 4.5, outside this fragment's scope. The companion `source_extracts/verify4c.py` checks spin/Euler/CG/Cartesian matrices, CHSH, partial trace and swapping exactly with SymPy (powers of $\hbar$ factored out), and checks moment formulas in $l = 0, \dots, 4$ representations; the general- l proofs are given above.